|INCH-POUND|

MIL-M-28787/527
29 DECEMBER 1989

~~MILITARY SPECIFICATION~~

MODULES, STANDARD ELECTRONIC

DIRECT MEMORY ACCESS II MODULE, KEY CODE DWH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Direct Memory Access (DMA) II module. This module shall conform to the class I requirements as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the ~~Department of Defense~~ Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.216 failures per million operating hours for class I when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DWH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C		60 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D2,D3	Ibd	V	S	0.0	1.4	0.0	1.4	0.0	1.4	mA
D4	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D5	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D6	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D7	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D8-D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D12	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D14	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D15	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
E1	Ios	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E2	Ioe	VI	S	-9.1	-6.91	-9.1	-6.91	-9.1	-6.91	mA
E3	Ioe	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E4,E5	Ios	VI	S	-9.1	-6.91	-9.1	-6.91	-9.1	-6.91	mA
E6	Ios	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E7	Ioe	VI	S	-9.1	-6.91	-9.1	-6.91	-9.1	-6.91	mA
E8	Ioe	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E9	Ios	VI	S	-9.1	-6.91	-9.1	-6.91	-9.1	-6.91	mA
E10	Ios	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E11	Ioe	VI	S	-9.1	-6.91	-9.1	-6.91	-9.1	-6.91	mA
E12	Ioe	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E13-E15	Ios	VI	S	-9.1	-6.91	-9.1	-6.91	-9.1	-6.91	mA
E16	Ios	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E17	Ioe	VI	S	-16.9	-6.91	-16.9	-6.91	-16.9	-6.91	mA
E18	Ioe	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 3/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C		60 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E19	Ioe	VI	S	-14.9	-6.91	-14.9	-6.91	-14.9	-6.91	mA
E20	Ios	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E21	Ios	VI	S	-12.3	-6.91	-12.3	-6.91	-12.3	-6.91	mA
E22	Ios	VI	S	-9.7	-6.91	-9.7	-6.91	-9.7	-6.91	mA
E23	Ioe	VI	S	-225.0	-35.6	-225.0	-35.6	-225.0	-35.6	mA
E24	Ios	VI	S	-9.7	-6.91	-9.7	-6.91	-9.7	-6.91	mA
E25-E39	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E40	Ioe	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E41-E44	Ioe	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E45	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E46-E48	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E49-E52	Ioe	VI	S	-225.0	-100.0	-225.0	-100.0	-225.0	-100.0	mA
E53	Ioe	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E54-E56	Ioe	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1,F2	Iih	VII	X	-6.83	-2.75	-6.83	-2.75	-6.83	-2.75	mA
F3-F8	Iih	VII	X	-6.85	-2.73	-6.85	-2.73	-6.85	-2.73	mA
F9	Iih	VII	X	-6.83	-2.75	-6.83	-2.75	-6.83	-2.75	mA
F10	Iih	VII	X	-14.6	-2.54	-14.6	-2.54	-14.6	-2.54	mA
F11	Iih	VII	X	-12.6	-2.59	-12.6	-2.59	-12.6	-2.59	mA
F12	Iih	VII	X	-10.0	-2.66	-10.0	-2.66	-10.0	-2.66	mA
F13,F14	Iih	VII	X	-7.43	-2.73	-7.43	-2.73	-7.43	-2.73	mA
F15	Iih	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F16	Iih	VII	X	-6.0	0.13	-6.0	0.13	-6.0	0.13	mA
F17	Iih	VII	X	-5.4	0.13	-5.4	0.13	-5.4	0.13	mA
F18	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F19	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F20	IiH	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F21	Iih	VII	X	-2.7	0.06	-2.7	0.06	-2.7	0.06	mA
F22,F23	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F24	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F25	Iih	VII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
F26	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C		60 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F27	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F28	Iih	VII	X	-3.0	0.06	-3.0	0.06	-3.0	0.06	mA
F29	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G1,G2	Iil	VIII	X	-9.9	-4.1	-9.9	-4.1	-9.9	-4.1	mA
G3-G8	Iil	VIII	X	-10.0	-4.0	-10.0	-4.0	-10.0	-4.0	mA
G9	Iil	VIII	X	-10.0	-4.1	-10.0	-4.1	-10.0	-4.1	mA
G10	Iil	VIII	X	-17.7	-3.8	-17.7	-3.8	-17.7	-3.8	mA
G11	Iil	VIII	X	-15.7	-3.9	-15.7	-3.9	-15.7	-3.9	mA
G12	Iil	VIII	X	-13.1	-4.0	-13.1	-4.0	-13.1	-4.0	mA
G13,G14	Iil	VIII	X	-10.5	-4.0	-10.5	-4.0	-10.5	-4.0	mA
G15	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G16	Iil	VIII	X	-6.0	0.13	-6.0	0.13	-6.0	0.13	mA
G17	Iil	VIII	X	-5.4	0.13	-5.4	0.13	-5.4	0.13	mA
G18	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G19	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G20	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G21	Iil	VIII	X	-2.7	0.06	-2.7	0.06	-2.7	0.06	mA
G22,G23	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G24	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G25	Iil	VIII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
G26	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G27	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G28	Iil	VIII	X	-3.0	0.06	-3.0	0.06	-3.0	0.06	mA
G29	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
H1-H32	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J65	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K20	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K21-K40	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 3°C 4/		0 ± 3°C		60 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L1-L9	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L10	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L15	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	7.4	13.8	7.4	13.8	7.4	13.8	uF
M2	PSFC	XIII	S	10.4	19.4	10.4	19.4	10.4	19.4	uF
M3	PSFC	XIII	S	2.3	4.3	2.3	4.3	2.3	4.3	uF
N1	Icc	XIV	X	400	980	400	960	400	960	mA
N2	Ioc	XIV	X	-65	-27	-65	-27	-65	-27	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R2	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R3	Ci/o	XVIII	Q	8	100	8	100	8	100	pF
R4,R5	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R6	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R7	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R8	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R9,R10	Ci/o	XVIII	Q	10	125	10	125	10	125	pF
R11-R14	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R15,R16	Ci/o	XVIII	Q	10	125	10	125	10	125	pF
R17	Ci/o	XVIII	Q	10	150	10	150	10	150	pF
R18	Ci/o	XVIII	Q	10	125	10	125	10	125	pF
R19,R20	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R21	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R22	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R23	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R24,R25	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R26	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R27	Ci/o	XVIII	Q	4	75	4	75	4	75	pF

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C		60 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R28	Ci/o	XVIII	Q	18	200	18	200	18	200	pF
R29	Ci/o	XVIII	Q	16	200	16	200	16	200	pF
R30	Ci/o	XVIII	Q	14	175	14	175	14	175	pF
R31	Ci/o	XVIII	Q	10	150	10	150	10	150	pF
R32,R33	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R34	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R35	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R36	Ci/o	XVIII	Q	10	125	10	125	10	125	pF
R37	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R38	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R39	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R40	Ci/o	XVIII	Q	4	50	4	50	4	50	pF
R41-R43	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R44,R45	Ci/o	XVIII	Q	10	125	10	125	10	125	pF
R46	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R47	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R48	Ci/o	XVIII	Q	6	75	6	75	6	75	pF
R49	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R50	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R51	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R52-R72	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R73	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R74	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R75	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R76	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R77	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
S	Vic latch up	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1-T3	PSTA	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/527-1 for class I. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc and (V2) of -16.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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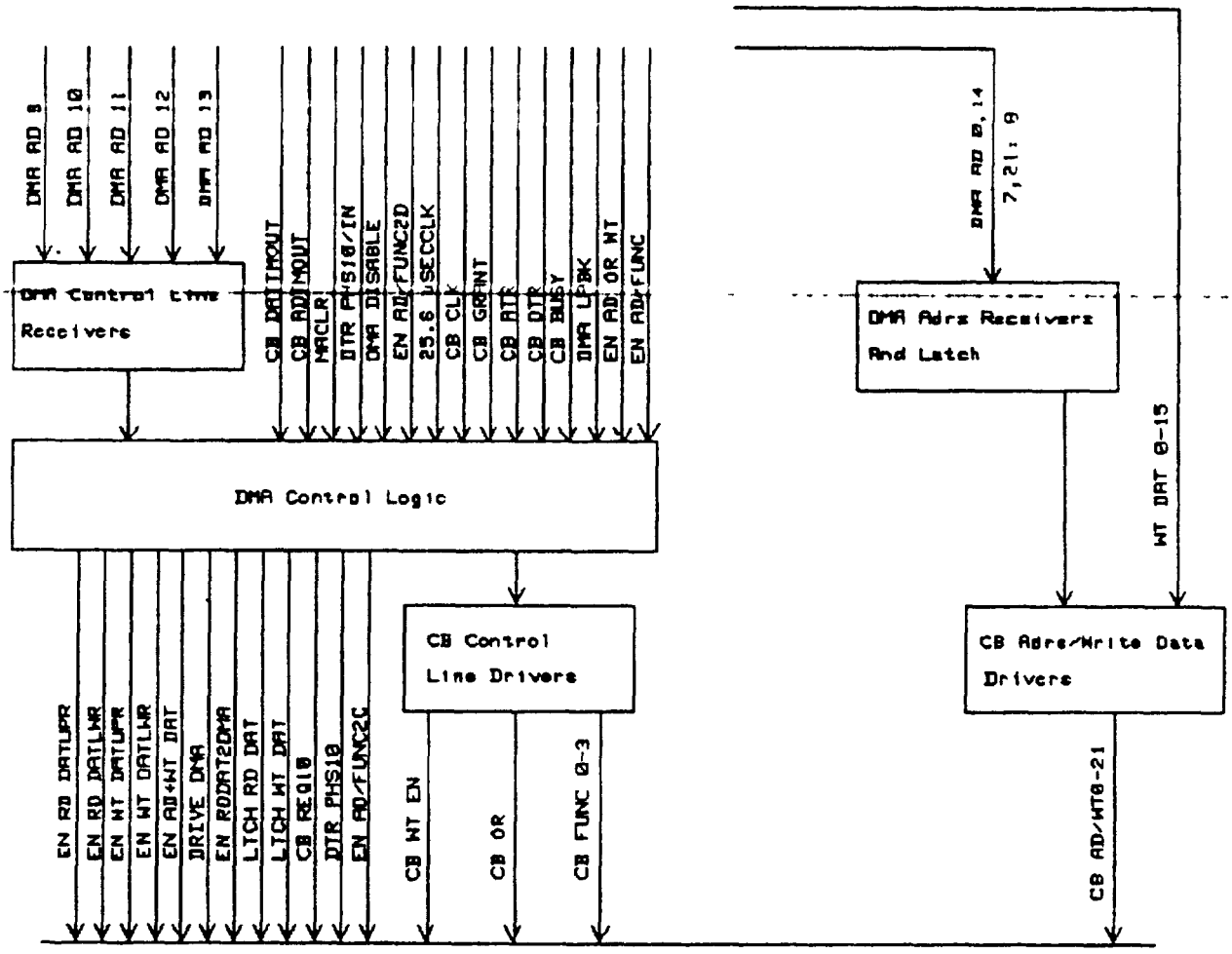


FIGURE 1. DWH functional block diagram.

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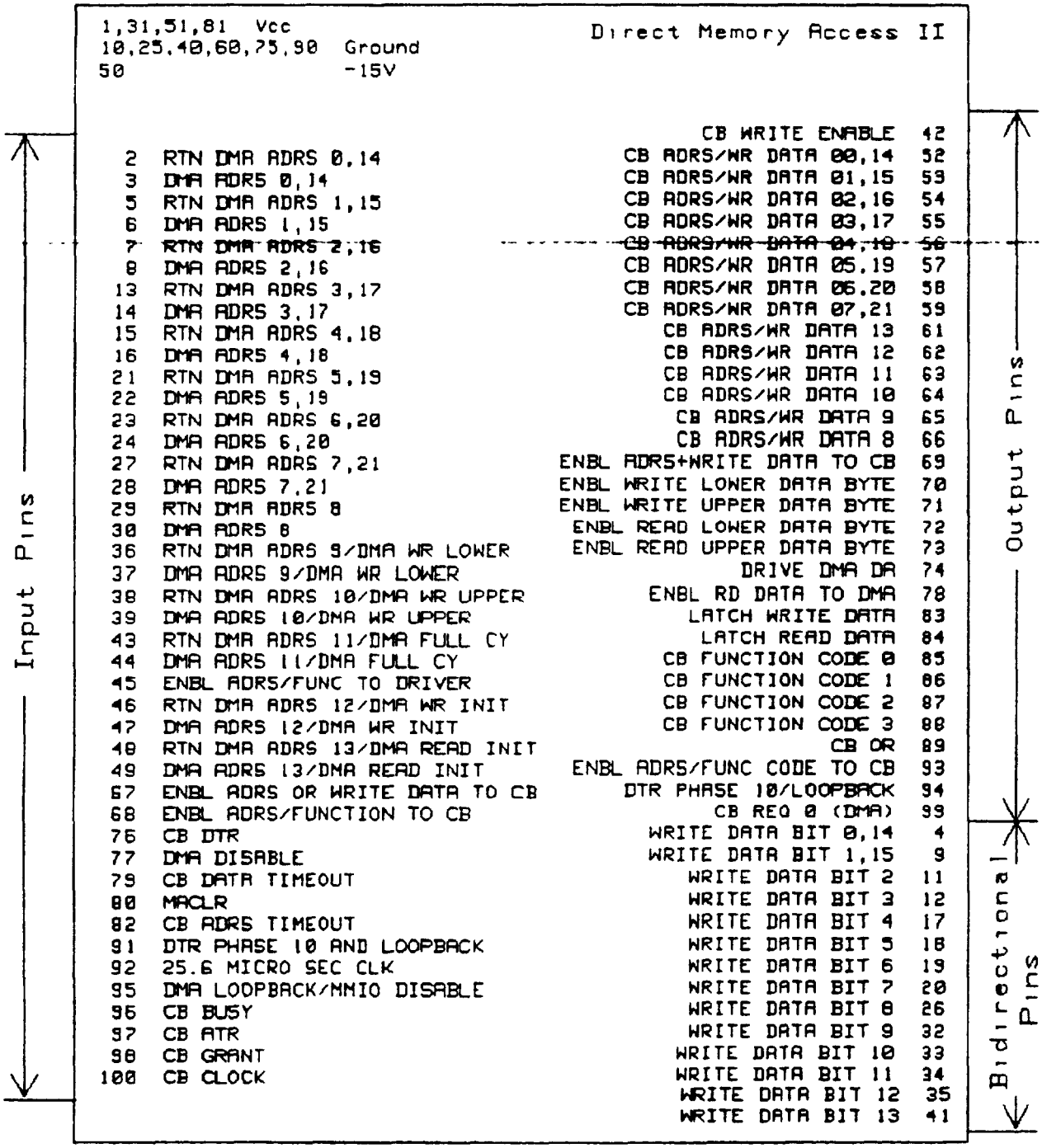


FIGURE 2. DWH pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall ~~be performed only at 25 ± 5°C.~~ If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

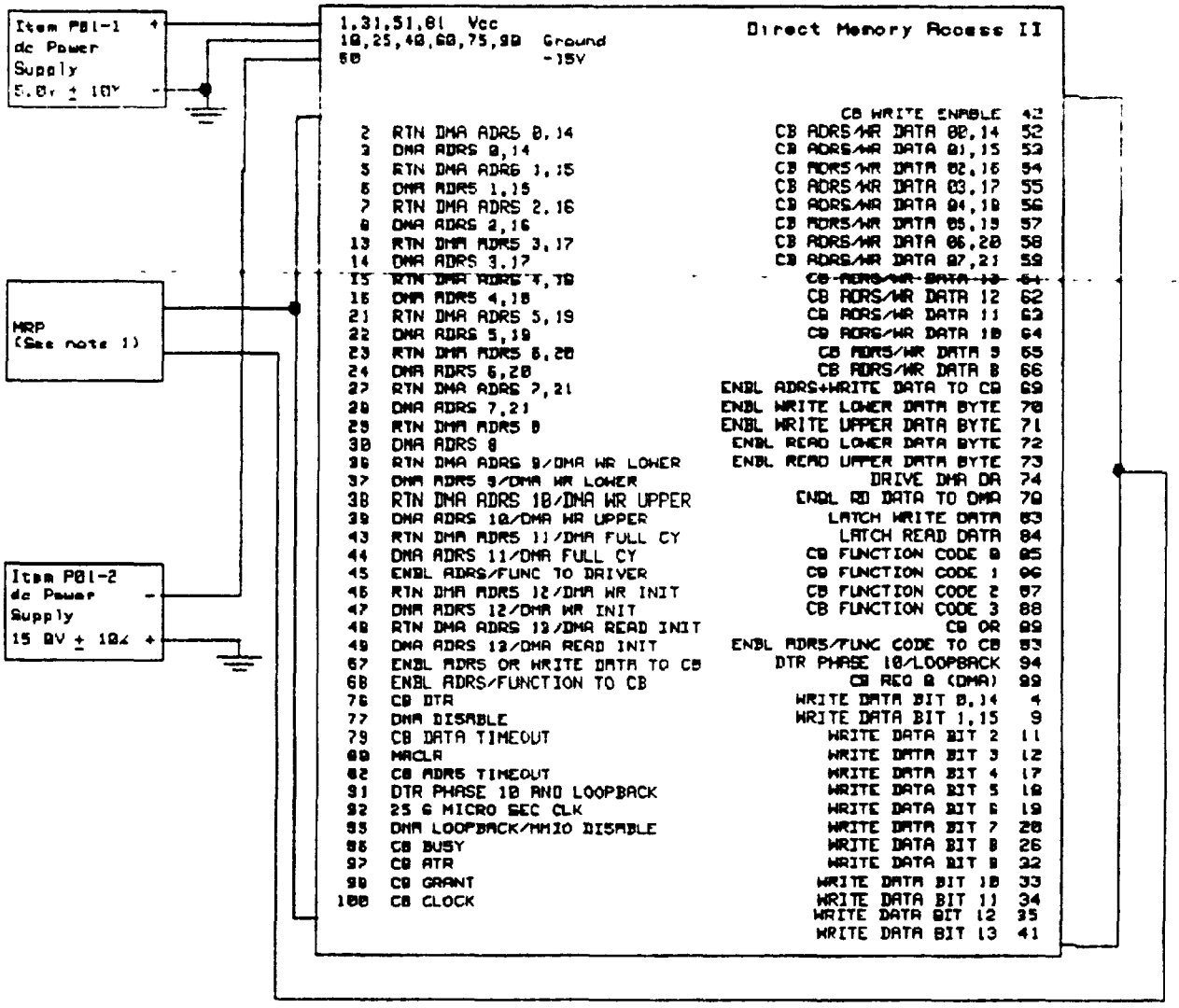
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. DWH operating temperature and component temperature test circuit.

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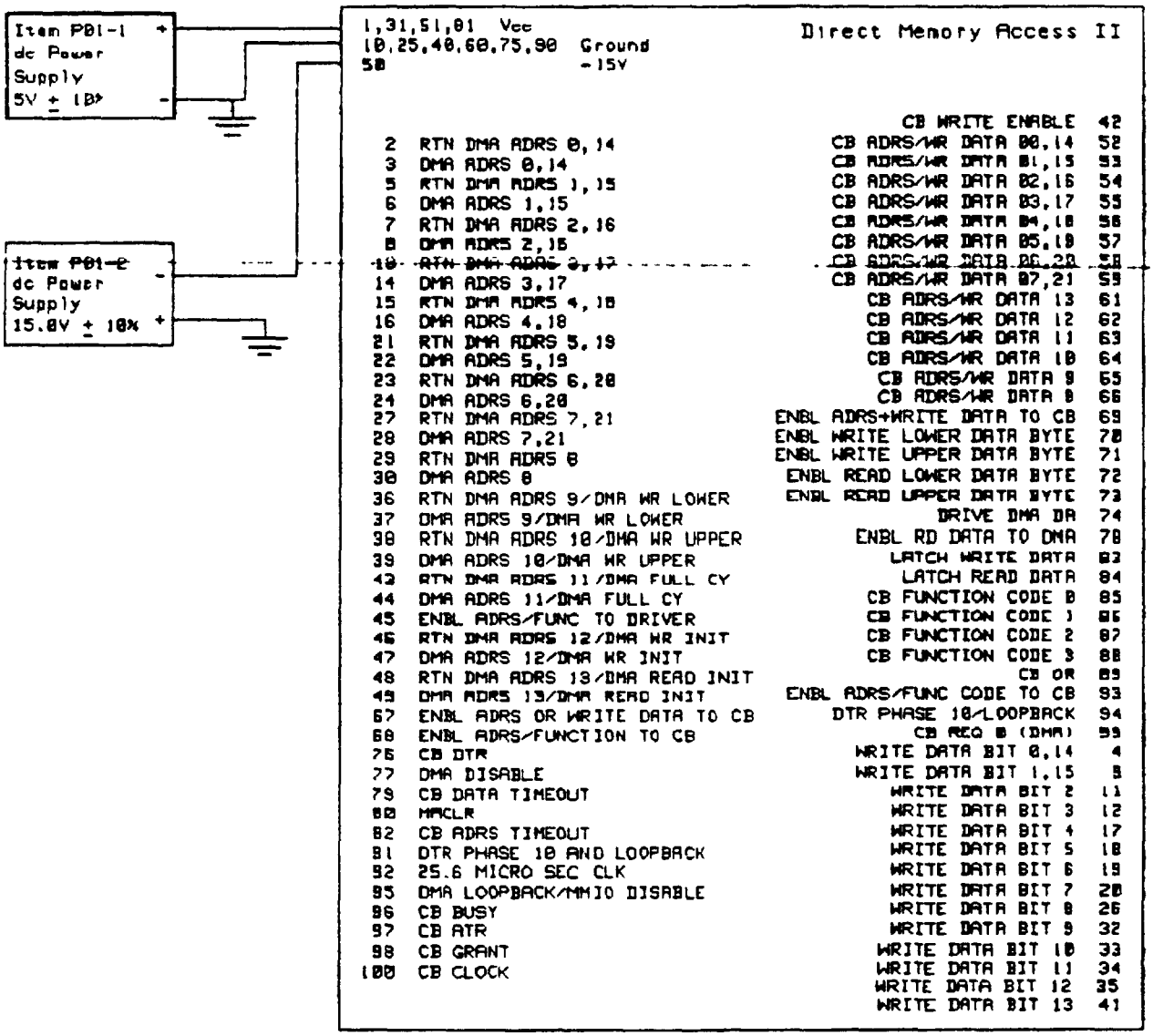


FIGURE 4. DWH vibration test circuit.

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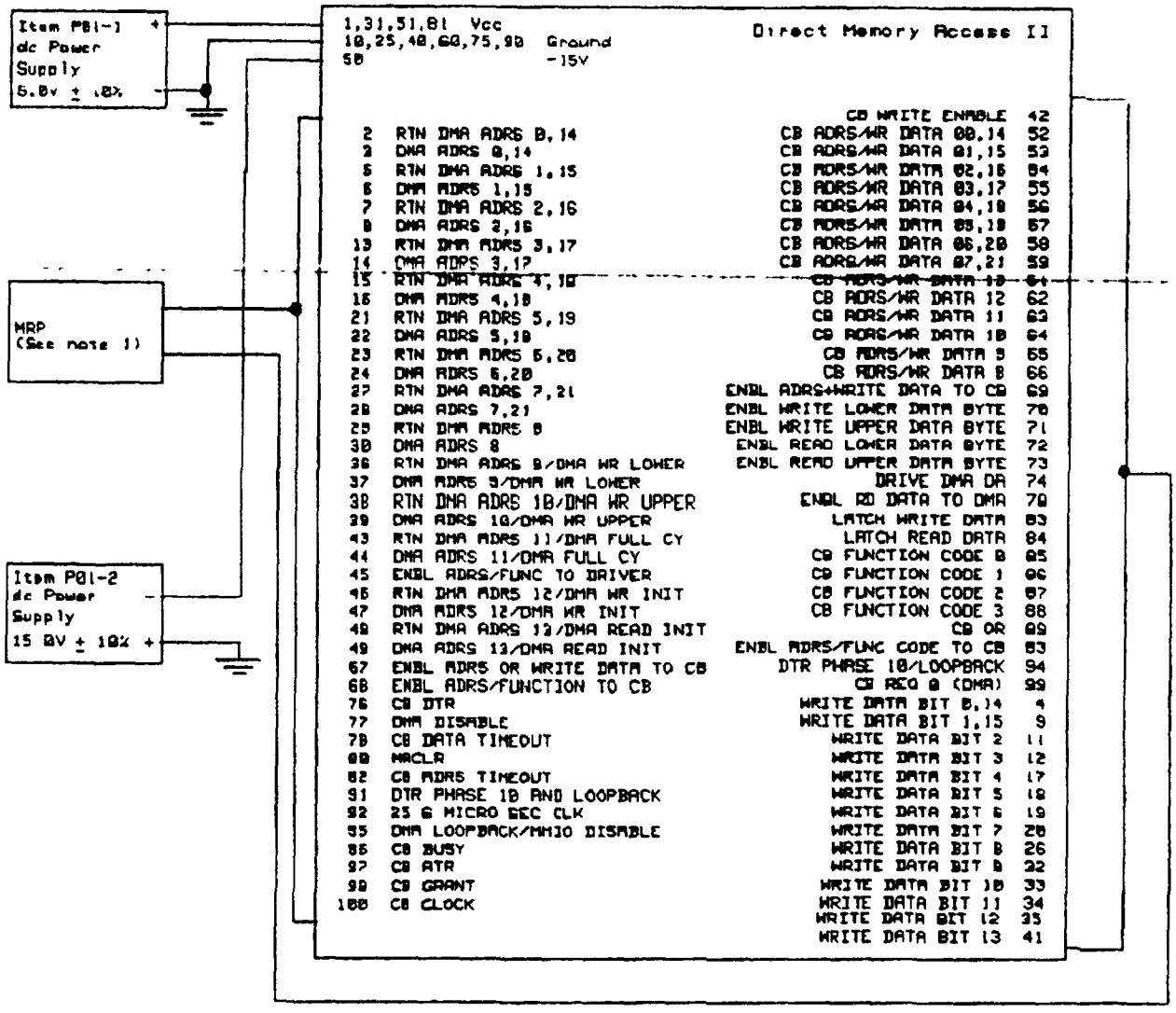
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. DWH life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 V dc to 5.5 V dc and (V2) -16.5 V dc to -13.5 V dc. Module power dissipation: 6.35 W (max); 3.89 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
WT DAT 0,14	NA	0.4	2.4	-9.9	-2.75	NA	74
WT DAT 1,15	NA	0.4	2.4	-9.9	-2.75	NA	78
WT DAT 2	NA	0.4	2.4	-10.0	-2.73	NA	91
WT DAT 3	NA	0.4	2.4	-10.0	-2.73	NA	93
WT DAT 4	NA	0.4	2.4	-10.0	-2.73	NA	100
WT DAT 5	NA	0.4	2.4	-10.0	-2.73	NA	86
WT DAT 6	NA	0.4	2.4	-10.0	-2.73	NA	103
WT DAT 7	NA	0.4	2.4	-10.0	-2.73	NA	90
WT DAT 8	NA	0.4	2.4	-10.0	-2.75	NA	81
WT DAT 9	NA	0.4	2.4	-17.7	-2.54	NA	173
WT DAT 10	NA	0.4	2.4	-15.7	-2.59	NA	198
WT DAT 11	NA	0.4	2.4	-13.1	-2.66	NA	144
WT DAT 12	NA	0.4	2.4	-10.5	-2.73	NA	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
WT DAT 13	NA	0.4	2.4	-10.5	-2.73	NA	100
EN AD/FUNC2D	-1.2	0.4	2.4	-2.6	0.07	1.1	55
EN AD OR WT	-1.2	0.4	2.4	-6.0	0.13	1.4	84
EN AD/FUNC	-1.2	0.4	2.4	-5.4	0.13	1.4	88
CB DTR	-1.2	0.4	2.4	-0.6	0.02	0.1	NA
DMA DISABLE	-1.2	0.4	2.4	-1.2	0.04	0.2	83
CB DATTMOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	28
MACLR	-1.2	0.4	2.4	-2.7	0.06	0.3	47
CB ADTMOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	22
DTR PHS10/IN	-1.2	0.4	2.4	-0.6	0.02	0.1	43
25.6 uSECCLK	-1.5	0.4	2.4	-0.4	0.02	0.1	19
DMA LPBK	-1.2	0.4	2.4	-1.6	0.02	0.1	38
CB BUSY	-1.2	0.4	2.4	-1.2	0.04	0.2	42

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB ATR	-1.2	0.4	2.4	-0.6	0.02	0.1	16
CB GRANT	-1.2	0.4	2.4	-3.0	0.06	0.3	71
CB CLK	-1.2	0.4	2.4	-4.0	0.1	1.0	14
RTN DMA 0,14	NA	NA	NA	NA	NA	NA	22
DMA AD 0,14	NA	0.4	2.4	NA	NA	NA	29
RTN DMA 1,15	NA	NA	NA	NA	NA	NA	34
DMA AD 1,15	NA	0.4	2.4	NA	NA	NA	37
RTN DMA 2,16	NA	NA	NA	NA	NA	NA	22
DMA AD 2,16	NA	0.4	2.4	NA	NA	NA	29
RTN DMA 3,17	NA	NA	NA	NA	NA	NA	34
DMA AD 3,17	NA	0.4	2.4	NA	NA	NA	38
RTN DMA 4,18	NA	NA	NA	NA	NA	NA	37
DMA AD 4,18	NA	0.4	2.4	NA	NA	NA	29

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RTN DMA 5,19	NA	NA	NA	NA	NA	NA	32
DMA AD 5,19	NA	0.4	2.4	NA	NA	NA	39
RTN DMA 6,20	NA	NA	NA	NA	NA	NA	23
DMA AD 6,20	NA	0.4	2.4	NA	NA	NA	29
RTN DMA 7,21	NA	NA	NA	NA	NA	NA	34
DMA AD 7,21	NA	0.4	2.4	NA	NA	NA	38
RTN DMA 8	NA	NA	NA	NA	NA	NA	24
DMA AD 8	NA	0.4	2.4	NA	NA	NA	32
RTN DMA 9	NA	NA	NA	NA	NA	NA	34
DMA AD 9	NA	0.4	2.4	NA	NA	NA	38
RTN DMA 10	NA	NA	NA	NA	NA	NA	22
DMA AD 10	NA	0.4	2.4	NA	NA	NA	29
RTN DMA 11	NA	NA	NA	NA	NA	NA	35

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DMA AD 11	NA	0.4	2.4	NA	NA	NA	23
RTN DMA 12	NA	NA	NA	NA	NA	NA	22
DMA AD 12	NA	0.4	2.4	NA	NA	NA	29
RTN DMA 13	NA	NA	NA	NA	NA	NA	36
DMA AD 13	NA	0.4	2.4	NA	NA	NA	37

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
WT DAT 0,14	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	74
WT DAT 1,15	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	78
WT DAT 2	NA	0.5	NA	1.6	NA	NA	-6.91	91
WT DAT 3	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	93
WT DAT 4	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	100
WT DAT 5	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	86
WT DAT 6	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	103
WT DAT 7	NA	0.5	NA	1.6	NA	NA	-6.91	90
WT DAT 8	NA	0.5	NA	1.6	NA	NA	-6.91	81
WT DAT 9	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	173
WT DAT 10	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	198
WT DAT 11	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	144
WT DAT 12	NA	0.5	NA	1.6	NA	NA	-6.91	125

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
WT DAT 13	NA	0.5	NA	1.6	NA	NA	-36.6 -6.91	100
CB WT EN	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	34
CB AD/WT0,14	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	30
CB AD/WT1,15	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	27
CB AD/WT2,16	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	26
CB AD/WT3,17	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	42
CB AD/WT4,18	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	27
CB AD/WT5,19	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	27
CB AD/WT6,20	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	29
CB AD/WT7,21	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	28
CB AD/WT13	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	40
CB AD/WT12	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	38
CB AD/WT11	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	35

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB AD/WT10	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	36
CB AD/WT9	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	31
CB AD/WT8	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	29
EN AD+WT DAT	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
EN WT DATLWR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
EN WT DATUPR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
EN RD DATLWR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
EN RD DATUPR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DRIVE DMA	2.4	0.5	-0.4	6.0	NA	NA	-100.0	NA
EN RDDAT2DMA	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LTCH WT DAT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LTCH RD DAT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CB FUNCT 0	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	33

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB FUNCT 1	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	32
CB FUNCT 2	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	32
CB FUNCT 3	2.4	NA	-0.4	NA	0.05	0.05	-100.0	33
CB OR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	36
EN AD/FUNC2C	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DTR PHS10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CB REQ10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to CB OR	-	-	-	36	36	36	nS	11
2	tpd	CB CLK to EN AD+WT DAT	-	-	-	38	38	38	nS	11
3	tpd	CB CLK to DTR PHS10	-	-	-	37	37	37	nS	11
4	tpd	CB CLK to EN AD/FUNC2C	-	-	-	38	38	38	nS	11
5	tpd	CB CLK to LTCH RD DAT	-	-	-	41	41	41	nS	11
6	tpd	CB CLK to DRIVE DMA	-	-	-	47	47	47	nS	11
7	tpd	CB CLK to EN WT DATLWR	-	-	-	48	48	48	nS	11
7	tpd	CB CLK to EN WT DATUPR	-	-	-	48	48	48	nS	11
8	tpd	CB CLK to EN RD DATLWR	-	-	-	48	48	48	nS	11
8	tpd	CB CLK to EN RD DATUPR	-	-	-	48	48	48	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	CB CLK to LTCH WT DAT	-	-	-	55	55	55	nS	11
10	tpd	CB CLK to CB WT EN	-	-	-	62	62	62	nS	11
11	tpd	DMA AD 0,14 to CB AD/WT0,14	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 1,15 to CB AD/WT1,15	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 2,16 to CB AD/WT2,16	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 3,17 to CB AD/WT3,17	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 4,18 to CB AD/WT4,18	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 5,19 to CB AD/WT5,19	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 6,20 to CB AD/WT6,20	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 7,21 to CB AD/WT7,21	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 8 to CB AD/WT8	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 9 to CB AD/WT9	-	-	-	91	91	91	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	DMA AD 10 to CB AD/WT10	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 11 to CB AD/WT11	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 12 to CB AD/WT12	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 13 to CB AD/WT13	-	-	-	91	91	91	nS	11

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6.4 Module function. DMA is an optional feature that allows an external device access to main memory. It is composed of two data handling modules (DMA I, DWG) and two address/control modules (DMA II, DWH). The DMA II modules contain receivers for DMA address bits 21-0 and control logic to direct the transfers and interface the DMA channel to the common bus. The control logic and common bus interface logic is active only on the DMA II upper module.

6.4.1 DMA operation. Three examples of DMA operation are described:

(a) DMA read operation.

(b) DMA write operation.

(c) DMA read/modify/write operation.

6.4.1.1 DMA read operations. DMA read operations are initiated by the external device setting the Read Initiate line and activating the DMA control lines. When the DMA control logic on the DMA II module receives the Read Initiate, a request is made for common bus usage. The bus arbitrator module responds with the Grant signal. The DMA II module then generates Originator Ready (OR) and places the common bus function and address on the lines. Memory interface sends an Addressed Terminator Ready (ATR), accepts the address and function bits and generates the Data Terminator Ready (DTR). At this time, data is available from memory. The data is latched at the DMA I Module and is available to the DMA device via the DMA bus when Data Available is active.

6.4.1.2 DMA write operations. DMA write operations are initiated by the external device setting the Read Initiate and DMA control lines applicable to the operation. When the DMA control logic receives Read Initiate, a common bus request is generated. The grant signal is received and OR, address, and function bits are placed on the lines. The ATR is received by the DMA II module and the address/function is accepted by the memory interface module. DMA data is placed on the common bus and is sampled when the second OR is generated by the DMA II module (after DTR generation). Data Available acts as an acknowledge to the external device to clear the lines when the cycle is completed.

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6.4.1.3 DMA read/modify/write operations. DMA read/modify/write operations are initiated by the external device setting the Read Initiate line and DMA control lines. The DMA control logic receives the Read Initiate and a common bus request is initiated. Operation then proceeds per the read cycle previously discussed. Modifications are completed and the second part of the cycle is initiated by the DMA control lines. With the exception of the Write Initiate signal which initiates the ~~write portion of the read/modify/write cycle, the balance of the~~ write cycle is similar to the write cycle previously discussed. Data Available acts an an acknowledge to the external device.

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6.4.1.5 Description of I/O pins.

Signal name	Pin number	I/O	Description
WT DAT 0,14	4	I/O	WRITE DATA BIT 0,14
WT DAT 1,15	9	I/O	WRITE DATA BIT 1,15
WT DAT 2	11	I/O	WRITE DATA BIT 2
WT DAT 3	12	I/O	WRITE DATA BIT 3
WT DAT 4	17	I/O	WRITE DATA BIT 4
WT DAT 5	18	I/O	WRITE DATA BIT 5
WT DAT 6	19	I/O	WRITE DATA BIT 6
WT DAT 7	20	I/O	WRITE DATA BIT 7
WT DAT 8	26	I/O	WRITE DATA BIT 8
WT DAT 9	32	I/O	WRITE DATA BIT 9
WT DAT 10	33	I/O	WRITE DATA BIT 10
WT DAT 11	34	I/O	WRITE DATA BIT 11
WT DAT 12	35	I/O	WRITE DATA BIT 12
WT DAT 13	41	I/O	WRITE DATA BIT 13
EN AD/FUNC2D	45	I	ENBL ADRS/FUNC TO DRIVER
EN AD OR WT	67	I	ENBL ADRS OR WRITE DATA TO CB
EN AD/FUNC	68	I	ENBL ADRS/FUNCTION TO CB
CB DTR	76	I	CB DTR
DMA DISABLE	77	I	DMA DISABLE
CB DATTMOUT	79	I	CB DATA TIMEOUT

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6.4.1.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MACLR	80	I	MACLR
CB ADTMOUT	82	I	CB ADRS TIMEOUT
DTR PHS10/IN	91	I	DTR PHASE 10 AND LOOPBACK
25.6 uSECCLK	92	I	25.6 MICRO SEC CLK
DMA LPBK	95	I	DMA LOOPBACK/MMIO DISABLE
CB BUSY	96	I	CB BUSY
CB ATR	97	I	CB ATR
CB GRANT	98	I	CB GRANT
CB CLK	100	I	CB CLOCK
CB WT EN	42	O	CB WRITE ENABLE
CB AD/WT0,14	52	O	CB ADRS/WR DATA 00,14
CB AD/WT1,15	53	O	CB ADRS/WR DATA 01,15
CB AD/WT2,16	54	O	CB ADRS/WR DATA 02,16
CB AD/WT3,17	55	O	CB ADRS/WR DATA 03,17
CB AD/WT4,18	56	O	CB ADRS/WR DATA 04,18
CB AD/WT5,19	57	O	CB ADRS/WR DATA 05,19
CB AD/WT6,20	58	O	CB ADRS/WR DATA 06,20
CB AD/WT7,21	59	O	CB ADRS/WR DATA 07,21
CB AD/WT13	61	O	CB ADRS/WR DATA 13
CB AD/WT12	62	O	CB ADRS/WR DATA 12

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6.4.1.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB AD/WT11	63	O	CB ADRS/WR DATA 11
CB AD/WT10	64	O	CB ADRS/WR DATA 10
CB AD/WT9	65	O	CB ADRS/WR DATA 09
CB AD/WT8	66	O	CB ADRS/WR DATA 08
EN AD+WT DAT	69	O	ENBL ADRS+WRITE DATA TO CB
EN WT DATLWR	70	O	ENBL WRITE LOWER DATA BYTE
EN WT DATUPR	71	O	ENBL WRITE UPPER DATA BYTE
EN RD DATLWR	72	O	ENBL READ LOWER DATA BYTE
EN RD DATUPR	73	O	ENBL READ UPPER DATA BYTE
DRIVE DMA	74	O	DRIVE DMA DA
EN RDDAT2DMA	78	O	ENBL RD DATA TO DMA
LTCH WT DAT	83	O	LATCH WRITE DATA
LTCH RD DAT	84	O	LATCH READ DATA
CB FUNCT 0	85	O	CB FUNCTION CODE 0
CB FUNCT 1	86	O	CB FUNCTION CODE 1
CB FUNCT 2	87	O	CB FUNCTION CODE 2
CB FUNCT 3	88	O	CB FUNCTION CODE 3
CB OR	89	O	CB OR
EN AD/FUNC2C	93	O	ENBL ADRS/FUNC CODE TO CB
DTR PHS10	94	O	DTR PHASE 10/LOOPBACK

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6.4.1.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB REQ10	99	O	CB REQ 0 (DMA)
RTN DMA 0,14	2	I	RTN DMA ADRS 0,14
DMA AD 0,14	3	I	DMA ADRS 0,14
RTN DMA 1,15	5	I	RTN DMA ADRS 1,15
DMA AD 1,15	6	I	DMA ADRS 1,15
RTN DMA 2,16	7	I	RTN DMA ADRS 2,16
DMA AD 2,16	8	I	DMA ADRS 2,16
RTN DMA 3,17	13	I	RTN DMA ADRS 3,17
DMA AD 3,17	14	I	DMA ADRS 3,17
RTN DMA 4,18	15	I	RTN DMA ADRS 4,18
DMA AD 4,18	16	I	DMA ADRS 4,18
RTN DMA 5,19	21	I	RTN DMA ADRS 5,19
DMA AD 5,19	22	I	DMA ADRS 5,19
RTN DMA 6,20	23	I	RTN DMA ADRS 6,20
DMA AD 6,20	24	I	DMA ADRS 6,20
RTN DMA 7,21	27	I	RTN DMA ADRS 7,21
DMA AD 7,21	28	I	DMA ADRS 7,21
RTN DMA 8	29	I	RTN DMA ADRS 8
DMA AD 8	30	I	DMA ADRS 8
RTN DMA 9	36	I	RTN DMA ADRS 9/DMA WR LOWER

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6.4.1.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DMA AD 9	37	I	DMA ADRS 9/DMA WR LOWER
RTN DMA 10	38	I	RTN DMA ADRS 10/DMA WR UPPER
DMA AD 10	39	I	DMA ADRS 10/DMA WR UPPER
RTN DMA 11	43	I	RTN DMA ADRS 11/DMA FULL CY
DMA AD 11	44	I	DMA ADRS 11/DMA FULL CY
RTN DMA 12	46	I	RTN DMA ADRS 12/DMA WR INIT
DMA AD 12	47	I	DMA ADRS 12/DMA WR INIT
RTN DMA 13	48	I	RTN DMA ADRS 13/DMA READ INIT
DMA AD 13	49	I	DMA ADRS 13/DMA READ INIT

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0047)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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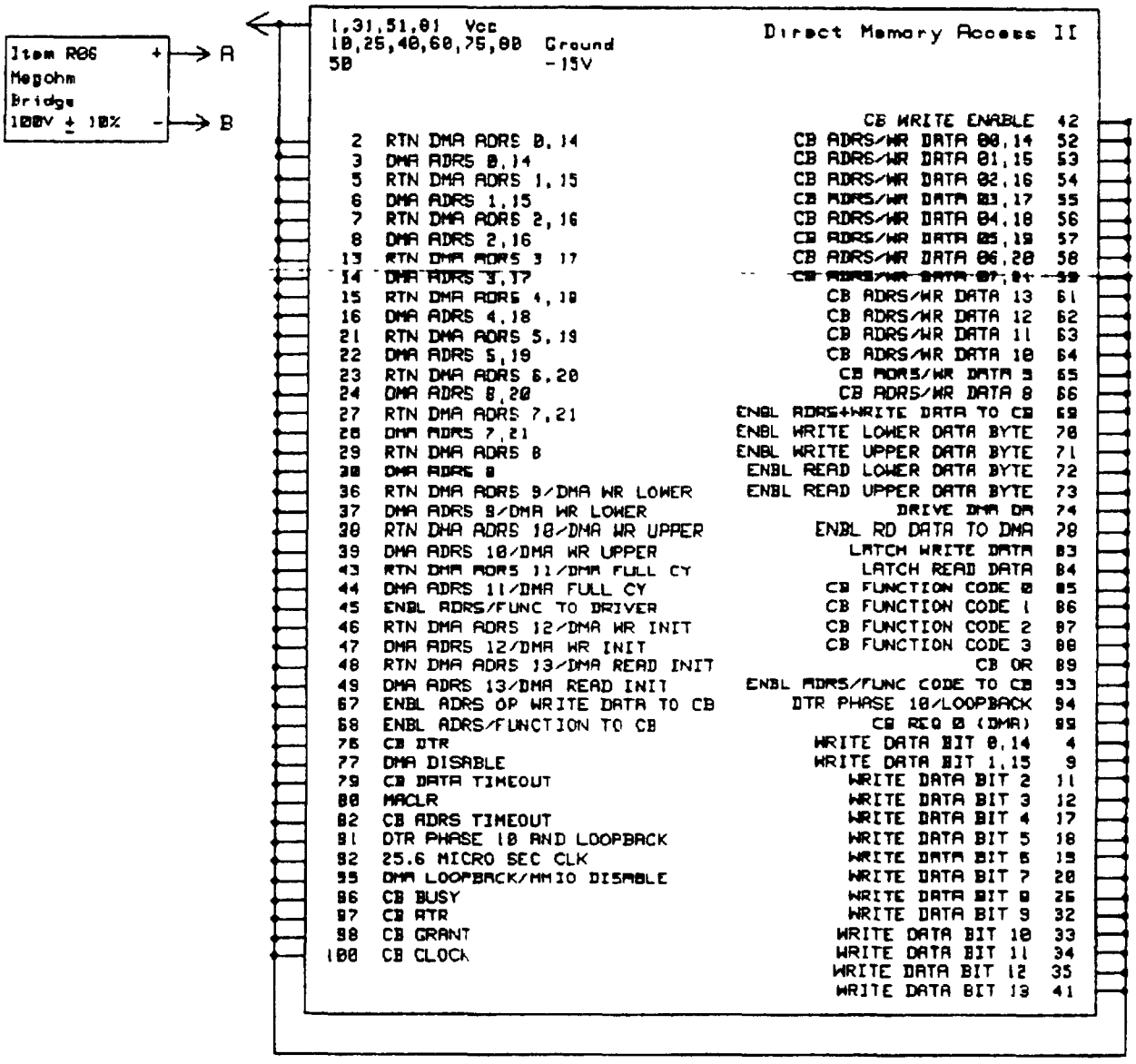


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 Vcc
10,25,40,60,75,90 Ground
50 -15V

Direct Memory Access II

		CB WRITE ENABLE	42
2	RTN DMA ADRS 0,14	CB ADRS/WR DATA 00,14	52
3	DMA ADRS 0,14	CB ADRS/WR DATA 01,15	53
5	RTN DMA ADRS 1,15	CB ADRS/WR DATA 02,16	54
6	DMA ADRS 1,15	CB ADRS/WR DATA 03,17	55
7	RTN DMA ADRS 2,16	CB ADRS/WR DATA 04,18	56
8	DMA ADRS 2,16	CB ADRS/WR DATA 05,19	57
10	RTN DMA ADRS 3,17	CB ADRS/WR DATA 06,20	58
14	DMA ADRS 3,17	CB ADRS/WR DATA 07,21	59
15	RTN DMA ADRS 4,18	CB ADRS/WR DATA 13	61
16	DMA ADRS 4,18	CB ADRS/WR DATA 12	62
21	RTN DMA ADRS 5,19	CB ADRS/WR DATA 11	63
22	DMA ADRS 5,19	CB ADRS/WR DATA 10	64
23	RTN DMA ADRS 6,20	CB ADRS/WR DATA 9	65
24	DMA ADRS 6,20	CB ADRS/WR DATA 8	66
27	RTN DMA ADRS 7,21	ENBL ADRS+WRITE DATA TO CB	69
28	DMA ADRS 7,21	ENBL WRITE LOWER DATA BYTE	70
29	RTN DMA ADRS 8	ENBL WRITE UPPER DATA BYTE	71
30	DMA ADRS 8	ENBL READ LOWER DATA BYTE	72
36	RTN DMA ADRS 9/DMA WR LOWER	ENBL READ UPPER DATA BYTE	73
37	DMA ADRS 9/DMA WR LOWER	DRIVE DMA DA	74
38	RTN DMA ADRS 10/DMA WR UPPER	ENBL RD DATA TO DMA	78
39	DMA ADRS 10/DMA WR UPPER	LATCH WRITE DATA	83
43	RTN DMA ADRS 11/DMA FULL CY	LATCH READ DATA	84
44	DMA ADRS 11/DMA FULL CY	CB FUNCTION CODE 0	85
45	ENBL ADRS/FUNC TO DRIVER	CB FUNCTION CODE 1	86
46	RTN DMA ADRS 12/DMA WR INIT	CB FUNCTION CODE 2	87
47	DMA ADRS 12/DMA WR INIT	CB FUNCTION CODE 3	88
48	RTN DMA ADRS 13/DMA READ INIT	CB OR	89
49	DMA ADRS 13/DMA READ INIT	ENBL ADRS/FUNC CODE TO CB	93
67	ENBL ADRS OR WRITE DATA TO CB	DTR PHASE 10/LOOPBACK	94
68	ENBL ADRS/FUNCTION TO CB	CB REQ 0 (DMA)	99
76	CB DTR	WRITE DATA BIT 0,14	4
77	DMA DISABLE	WRITE DATA BIT 1,15	9
79	CB DATA TIMEOUT	WRITE DATA BIT 2	11
80	MRCLR	WRITE DATA BIT 3	12
82	CB ADRS TIMEOUT	WRITE DATA BIT 4	17
91	DTR PHASE 10 AND LOOPBACK	WRITE DATA BIT 5	18
92	25 6 MICRO SEC CLK	WRITE DATA BIT 6	19
95	DMA LOOPBACK/MMIO DISABLE	WRITE DATA BIT 7	20
96	CB BUSY	WRITE DATA BIT 8	26
97	CB ATR	WRITE DATA BIT 9	32
98	CB GRANT	WRITE DATA BIT 10	33
100	CB CLOCK	WRITE DATA BIT 11	34
		WRITE DATA BIT 12	35
		WRITE DATA BIT 13	41

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V2	17.5	10,25, 40,60, 75,90	50	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
(V2)	-17.5 V	+175 mV,- 0mV	C
Power supply current limit, (Vcc)	1.73 A	NA	C
(V2)	-0.12 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

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NOTES: - Continued.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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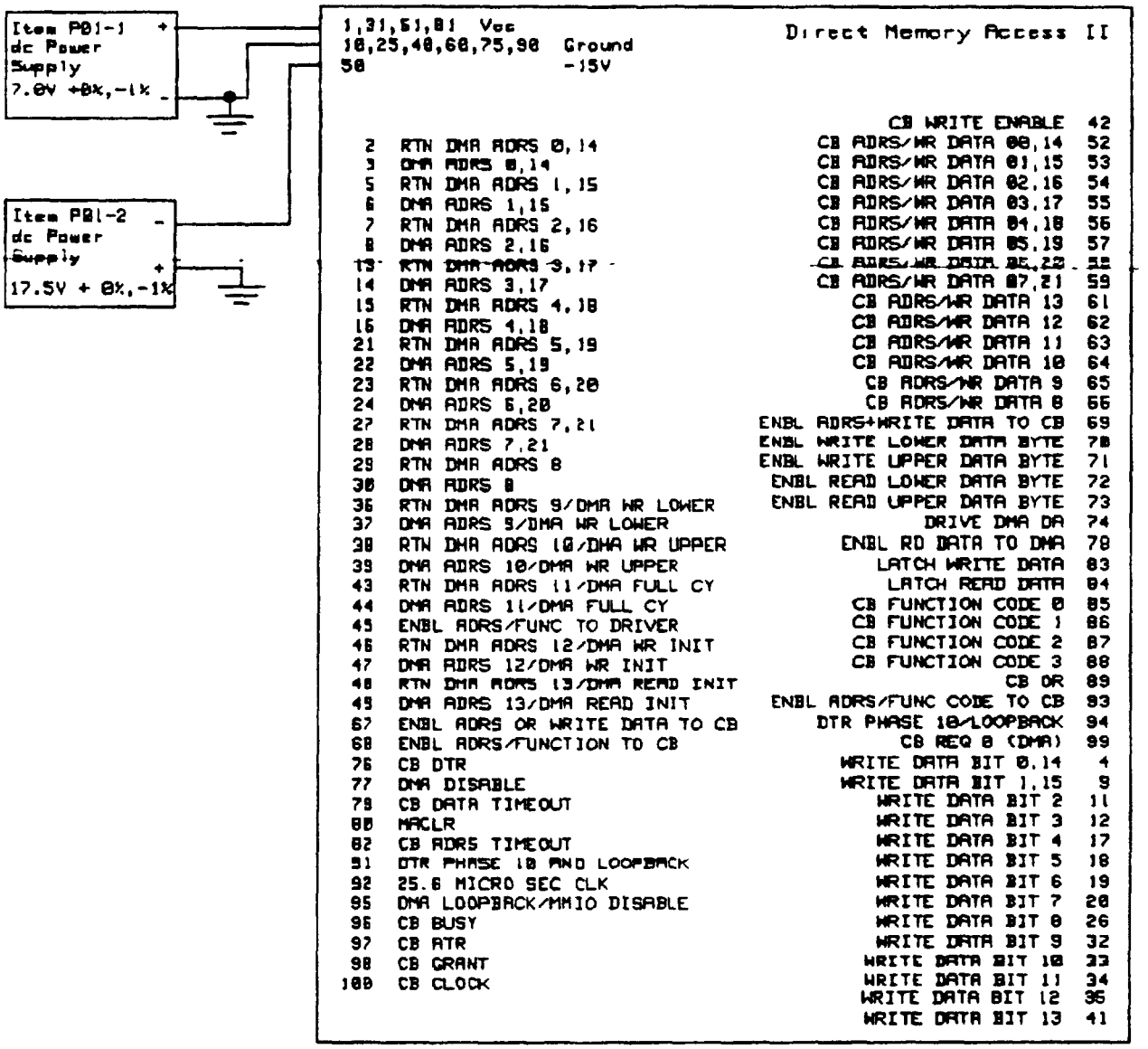


FIGURE 8. Power supply stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	EN AD/FUNC2D	45	5.5	0.0	1.1	mA	1-3
D2	EN AD OR WT	67	5.5	0.0	1.4	mA	1-3
D3	EN AD/FUNC	68	5.5	0.0	1.4	mA	1-3
D4	CB DTR	76	5.5	0.0	0.1	mA	1-3
D5	DMA DISABLE	77	5.5	0.0	0.2	mA	1-3
D6	CB DATTMOUT	79	5.5	0.0	0.1	mA	1-3
D7	MACLR	80	5.5	0.0	0.3	mA	1-3
D8	CB ADTMOUT	82	5.5	0.0	0.1	mA	1-3
D9	DTR PHS10/IN	91					
D10	25.6 uSECCLK	92					
D11	DMA LPBK	95	5.5	0.0	0.1	mA	1-3
D12	CB BUSY	96	5.5	0.0	0.2	mA	1-3
D13	CB ATR	97	5.5	0.0	0.1	mA	1-3
D14	CB GRANT	98	5.5	0.0	0.3	mA	1-3
D15	CB CLK	100	5.5	0.0	1.0	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D15
(V2)	-16.5 V	± 165 mV	D1-D15
Power supply current limit, (Vcc)	1.44 A	NA	D1-D15
(V2)	-0.10 A	NA	D1-D15
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D15
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D15
low, (VL)	0.0 V	± 100 mV	D1-D15
Input current measurement	NA	± 55 uA	D1
	NA	± 70 uA	D2,D3
	NA	± 5 uA	D4
	NA	± 10 uA	D5
	NA	± 5 uA	D6
	NA	± 15 uA	D7
	NA	± 5 uA	D8-D11
	NA	± 10 uA	D12
	NA	± 5 uA	D13
	NA	± 15 uA	D14
	NA	± 50 uA	D15

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	WT DAT 0,14	4	0.0	-225.0	-35.6	mA	1-4
E2	WT DAT 0,14	4	0.0	-9.1	-6.91	mA	1-4
E3	WT DAT 1,15	9	0.0	-225.0	-35.6	mA	1-4
E4	WT DAT 1,15	9	0.0	-9.1	-6.91	mA	1-4
E5	WT DAT 2	11	0.0	-9.1	-6.91	mA	1-4
E6	WT DAT 3	12	0.0	-225.0	-35.6	mA	1-4
E7	WT DAT 3	12	0.0	-9.1	-6.91	mA	1-4
E8	WT DAT 4	17	0.0	-225.0	-35.6	mA	1-4
E9	WT DAT 4	17	0.0	-9.1	-6.91	mA	1-4
E10	WT DAT 5	18	0.0	-225.0	-35.6	mA	1-4
E11	WT DAT 5	18	0.0	-9.1	-6.91	mA	1-4
E12	WT DAT 6	19	0.0	-225.0	-35.6	mA	1-4
E13	WT DAT 6	19	0.0	-9.1	-6.91	mA	1-4
E14	WT DAT 7	20					
E15	WT DAT 8	26	0.0	-9.1	-6.91	mA	1-4
E16	WT DAT 9	32	0.0	-225.0	-35.6	mA	1-4
E17	WT DAT 9	32	0.0	-16.9	-6.91	mA	1-4
E18	WT DAT 10	33	0.0	-225.0	-35.6	mA	1-4
E19	WT DAT 10	33	0.0	-14.9	-6.91	mA	1-4
E20	WT DAT 11	34	0.0	-225.0	-35.6	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	WT DAT 11	34	0.0	-12.3	-6.91	mA	1-4
E22	WT DAT 12	35	0.0	-9.7	-6.91	mA	1-4
E23	WT DAT 13	41	0.0	-225.0	-35.6	mA	1-4
E24	WT DAT 13	41	0.0	-9.7	-6.91	mA	1-4
E25	CB WT EN	42	0.0	-325.0	-100.0	mA	1-4
E26	CB AD/WT0,14	52					
E27	CB AD/WT1,15	53					
E28	CB AD/WT2,16	54					
E29	CB AD/WT3,17	55					
E30	CB AD/WT4,18	56					
E31	CB AD/WT5,19	57					
E32	CB AD/WT6,20	58					
E33	CB AD/WT7,21	59					
E34	CB AD/WT13	61					
E35	CB AD/WT12	62					
E36	CB AD/WT11	63					
E37	CB AD/WT10	64					
E38	CB AD/WT9	65					
E39	CB AD/WT8	66	0.0	-325.0	-100.0	mA	1-4
E40	EN AD+WT DAT	69	0.0	-100.0	-40.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	EN WT DATLWR	70	0.0	-150.0	-60.0	mA	1-4
E42	EN WT DATUPR	71					
E43	EN RD DATLWR	72					
E44	EN RD DATUPR	73	0.0	-150.0	-60.0	mA	1-4
E45	DRIVE DMA	74	0.0	-325.0	-100.0	mA	1-4
E46	EN RDDAT2DMA	78	0.0	-150.0	-60.0	mA	1-4
E47	LTCH WT DAT	83					
E48	LTCH RD DAT	84	0.0	-150.0	-60.0	mA	1-4
E49	CB FUNCT 0	85	0.0	-225.0	-100.0	mA	1-4
E50	CB FUNCT 1	86					
E51	CB FUNCT 2	87					
E52	CB FUNCT 3	88	0.0	-225.0	-100.0	mA	1-4
E53	CB OR	89	0.0	-325.0	-100.0	mA	1-4
E54	EN AD/FUNC2C	93	0.0	-150.0	-60.0	mA	1-4
E55	DTR PHS10	94					
E56	CB REQ10	99	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E56
(V2)	-16.5 V	± 165 mV	E1-E56
Power supply current limit, (Vcc)	1.44 A	NA	E1-E56
(V2)	-0.10 A	NA	E1-E56
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E56
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E56
low, (VL)	0.0 V	± 100 mV	E1-E56
Output current measurement	NA	± 9.47 mA	E1
	NA	± 0.11 mA	E2
	NA	± 9.47 mA	E3
	NA	± 0.11 mA	E4,E5
	NA	± 9.47 mA	E6
	NA	± 0.11 mA	E7
	NA	± 9.47 mA	E8
	NA	± 0.11 mA	E9
	NA	± 9.47 mA	E10
	NA	± 0.11 mA	E11
	NA	± 9.47 mA	E12
	NA	± 0.11 mA	E13-E15
	NA	± 9.47 mA	E16
	NA	± 0.5 mA	E17
	NA	± 9.47 mA	E18
	NA	± 0.4 mA	E19

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 9.47 mA	E20
	NA	± 0.27 mA	E21
	NA	± 0.14 mA	E22
	NA	± 9.47 mA	E23
	NA	± 0.14 mA	E24
	NA	± 11.2 mA	E25-E39
	NA	± 3.00 mA	E40
	NA	± 4.5 mA	E41-E44
	NA	± 11.2 mA	E45
	NA	± 4.5 mA	E46-E48
	NA	± 6.25 mA	E49-E52
	NA	± 11.2 mA	E53
	NA	± 4.5 mA	E54-E56

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	WT DAT 0,14	4	2.4	-6.83	-2.75	mA	1-3
F2	WT DAT 1,15	9	2.4	-6.83	-2.75	mA	1-3
F3	WT DAT 2	11	2.4	-6.85	-2.73	mA	1-3
F4	WT DAT 3	12					
F5	WT DAT 4	17					
F6	WT DAT 5	18					
F7	WT DAT 6	19					
F8	WT DAT 7	20	2.4	-6.85	-2.73	mA	1-3
F9	WT DAT 8	26	2.4	-6.83	-2.75	mA	1-3
F10	WT DAT 9	32	2.4	-14.6	-2.54	mA	1-3
F11	WT DAT 10	33	2.4	-12.6	-2.59	mA	1-3
F12	WT DAT 11	34	2.4	-10.0	-2.66	mA	1-3
F13	WT DAT 12	35	2.4	-7.43	-2.73	mA	1-3
F14	WT DAT 13	41	2.4	-7.43	-2.73	mA	1-3
F15	EN AD/FUNC2D	45	2.4	-2.6	0.07	mA	1-3
F16	EN AD OR WT	67	2.4	-6.0	0.13	mA	1-3
F17	EN AD/FUNC	68	2.4	-5.4	0.13	mA	1-3
F18	CB DTR	76	2.4	-0.6	0.02	mA	1-3
F19	DMA DISABLE	77	2.4	-1.2	0.04	mA	1-3
F20	CB DATTMOUT	79	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	MACLR	80	2.4	-2.7	0.06	mA	1-3
F22	CB ADTMOUT	82	2.4	-0.6	0.02	mA	1-3
F23	DTR PHS10/IN	91	2.4	-0.6	0.02	mA	1-3
F24	25.6 uSECCLK	92	2.4	-0.4	0.02	mA	1-3
F25	DMA LPBK	95	2.4	-1.6	0.02	mA	1-3
F26	CB BUSY	96	2.4	-1.2	0.04	mA	1-3
F27	CB ATR	97	2.4	-0.6	0.02	mA	1-3
F28	CB GRANT	98	2.4	-3.0	0.06	mA	1-3
F29	CB CLK	100	2.4	-4.0	0.1	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F29
(V2)	-16.5 V	± 165 mV	F1-F29
Power supply current limit, (Vcc)	1.44 A	NA	F1-F29
(V2)	-0.10 A	NA	F1-F29
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F29
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F29
low, (VL)	0.0 V	± 100 mV	F1-F29
Input current measurement	NA	± 204 uA	F1,F2
	NA	± 206 uA	F3-F8
	NA	± 204 uA	F9
	NA	± 600 uA	F10
	NA	± 500 uA	F11
	NA	± 370 uA	F12
	NA	± 235 uA	F13,F14
	NA	± 133 uA	F15
	NA	± 306 uA	F16
	NA	± 277 uA	F17
	NA	± 31 uA	F18
	NA	± 62 uA	F19
	NA	± 31 uA	F20
	NA	± 138 uA	F21
	NA	± 31 uA	F22,F23
	NA	± 21 uA	F24

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 81 uA	F25
	NA	± 62 uA	F26
	NA	± 31 uA	F27
	NA	± 153 uA	F28
	NA	± 205 uA	F29

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	WT DAT 0,14	4	0.4	-9.9	-4.1	mA	1-3
G2	WT DAT 1,15	9	0.4	-9.9	-4.1	mA	1-3
G3	WT DAT 2	11	0.4	-10.0	-4.0	mA	1-3
G4	WT DAT 3	12					
G5	WT DAT 4	17					
G6	WT DAT 5	18					
G7	WT DAT 6	19					
G8	WT DAT 7	20	0.4	-10.0	-4.0	mA	1-3
G9	WT DAT 8	26	0.4	-10.0	-4.1	mA	1-3
G10	WT DAT 9	32	0.4	-17.7	-3.8	mA	1-3
G11	WT DAT 10	33	0.4	-15.7	-3.9	mA	1-3
G12	WT DAT 11	34	0.4	-13.1	-4.0	mA	1-3
G13	WT DAT 12	35	0.4	-10.5	-4.0	mA	1-3
G14	WT DAT 13	41	0.4	-10.5	-4.0	mA	1-3
G15	EN AD/FUNC2D	45	0.4	-2.6	0.07	mA	1-3
G16	EN AD OR WT	67	0.4	-6.0	0.13	mA	1-3
G17	EN AD/FUNC	68	0.4	-5.4	0.13	mA	1-3
G18	CB DTR	76	0.4	-0.6	0.02	mA	1-3
G19	DMA DISABLE	77	0.4	-1.2	0.04	mA	1-3
G20	CB DATTMOUT	79	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	MACLR	80	0.4	-2.7	0.06	mA	1-3
G22	CB ADTMOUT	82	0.4	-0.6	0.02	mA	1-3
G23	DTR PHS10/IN	91	0.4	-0.6	0.02	mA	1-3
G24	25.6 uSECCLK	92	0.4	-0.4	0.02	mA	1-3
G25	DMA LPBK	95	0.4	-1.6	0.02	mA	1-3
G26	CB BUSY	96	0.4	-1.2	0.04	mA	1-3
G27	CB ATR	97	0.4	-0.6	0.02	mA	1-3
G28	CB GRANT	98	0.4	-3.0	0.06	mA	1-3
G29	CB CLK	100	0.4	-4.0	0.1	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G29
(V2)	-16.5 V	± 165 mV	G1-G29
Power supply current limit, (Vcc)	1.44 A	NA	G1-G29
(V2)	-0.10 A	NA	G1-G29
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G29
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G29
low, (VL)	0.0 V	± 100 mV	G1-G29
Input current measurement	NA	± 290 uA	G1,G2
	NA	± 300 uA	G3-G8
	NA	± 295 uA	G9
	NA	± 700 uA	G10
	NA	± 590 uA	G11
	NA	± 460 uA	G12
	NA	± 330 uA	G13,G14
	NA	± 133 uA	G15
	NA	± 306 uA	G16
	NA	± 277 uA	G17
	NA	± 31 uA	G18
	NA	± 62 uA	G19
	NA	± 31 uA	G20
	NA	± 138 uA	G21
	NA	± 31 uA	G22,G23
	NA	± 21 uA	G24

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 81 uA	G25
	~~NA~~	~~± 62 uA~~	~~G26~~
	NA	± 31 uA	G27
	NA	± 153 uA	G28
	NA	± 205 uA	G29

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	CB WT EN	42	-0.4	2.4	4.5	V	1-4
H2	CB AD/WT0,14	52					
H3	CB AD/WT1,15	53					
H4	CB AD/WT2,16	54					
H5	CB AD/WT3,17	55					
H6	CB AD/WT4,18	56					
H7	CB AD/WT5,19	57					
H8	CB AD/WT6,20	58					
H9	CB AD/WT7,21	59					
H10	CB AD/WT13	61					
H11	CB AD/WT12	62					
H12	CB AD/WT11	63					
H13	CB AD/WT10	64					
H14	CB AD/WT9	65					
H15	CB AD/WT8	66					
H16	EN AD+WT DAT	69					
H17	EN WT DATLWR	70					
H18	EN WT DATUPR	71					
H19	EN RD DATLWR	72					
H20	EN RD DATUPR	73	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	DRIVE DMA	74	-0.4	2.4	4.5	V	1-4
H22	EN RDDAT2DMA	78					
H23	LTCH WT DAT	83					
H24	LTCH RD DAT	84					
H25	CB FUNCT 0	85					
H26	CB FUNCT 1	86					
H27	CB FUNCT 2	87					
H28	CB FUNCT 3	88					
H29	CB OR	89					
H30	EN AD/FUNC2C	93					
H31	DTR PHS10	94					
H32	CB REQ10	99	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H32
(V2)	-13.5 V	± 135 mV	H1-H32
Power supply current limit, (Vcc)	1.44 A	NA	H1-H32
(V2)	-0.10 A	NA	H1-H32
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H32
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H32
low, (VL)	0.0 V	± 100 mV	H1-H32
Output voltage measurement	NA	± 105 mV	H1-H32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

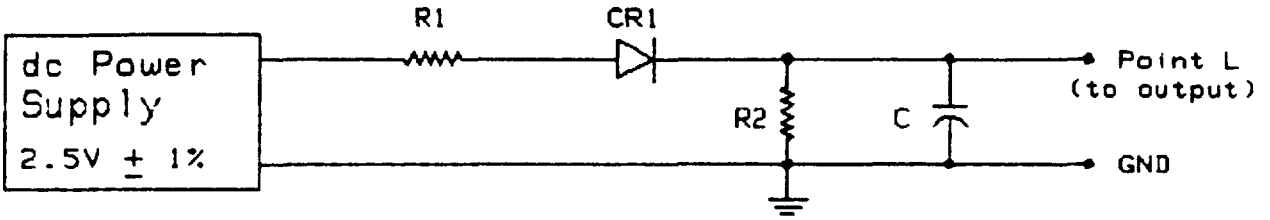
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
42,52,53,54,55,56, 57,58,59,61,62,63, 64,65,66,69,70,71, 72,73,74,78,83,84, 85,86,87,89,93,94, 99	6200	6.0	0.5
88	6200	NA	NA
4,9,11,12,17,18, 19,20,26,32,33,34, 35,41	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	WT DAT 0,14	4	1.6	0.0	0.5	V	1-4
J2	WT DAT 0,14	4					
J3	WT DAT 1,15	9					
J4	WT DAT 1,15	9					
J5	WT DAT 2	11					
J6	WT DAT 2	11					
J7	WT DAT 2	11					
J8	WT DAT 3	12					
J9	WT DAT 3	12					
J10	WT DAT 3	12					
J11	WT DAT 4	17					
J12	WT DAT 4	17					
J13	WT DAT 4	17					
J14	WT DAT 5	18					
J15	WT DAT 5	18					
J16	WT DAT 5	18					
J17	WT DAT 6	19					
J18	WT DAT 6	19					
J19	WT DAT 6	19					
J20	WT DAT 7	20	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	WT DAT 7	20	1.6	0.0	0.5	V	1-4
J22	WT DAT 7	20					
J23	WT DAT 8	26					
J24	WT DAT 8	26					
J25	WT DAT 9	32					
J26	WT DAT 9	32					
J27	WT DAT 10	33					
J28	WT DAT 10	33					
J29	WT DAT 11	34					
J30	WT DAT 11	34					
J31	WT DAT 12	35					
J32	WT DAT 12	35					
J33	WT DAT 13	41					
J34	WT DAT 13	41	1.6	0.0	0.5	V	1-4
J35	CB WT EN	42	6.0	0.0	0.5	V	1-4
J36	CB AD/WT0,14	52					
J37	CB AD/WT1,15	53					
J38	CB AD/WT2,16	54					
J39	CB AD/WT3,17	55					
J40	CB AD/WT4,18	56	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J41	CB AD/WT5,19	57	6.0	0.0	0.5	V	1-4
J42	CB AD/WT6,20	58					
J43	CB AD/WT7,21	59					
J44	CB AD/WT13	61					
J45	CB AD/WT12	62					
J46	CB AD/WT11	63					
J47	CB AD/WT10	64					
J48	CB AD/WT9	65					
J49	CB AD/WT8	66					
J50	EN AD+WT DAT	69					
J51	EN WT DATLWR	70					
J52	EN WT DATUPR	71					
J53	EN RD DATLWR	72					
J54	EN RD DATUPR	73					
J55	DRIVE DMA	74					
J56	EN RDDAT2DMA	78					
J57	LTCH WT DAT	83					
J58	LTCH RD DAT	84					
J59	CB FUNCT 0	85					
J60	CB FUNCT 1	86	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J61	CB FUNCT 2	87	6.0	0.0	0.5	V	1-4
J62	CB OR	89					
J63	EN AD/FUNC2C	93					
J64	DTR PHS10	94					
J65	CB REQ10	99	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	4.5 V -13.5 V	± 45 mV ± 135 mV	J1-J65 J1-J65
Power supply current limit, (Vcc) (V2)	1.44 A -0.10 A	NA NA	J1-J65 J1-J65
Forcing current, (Iol)	1.6 mA 6.0 mA	± 160 uA ± 600 uA	J1-J34 J35-J65
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J65 J1-J65
Output voltage measurement	NA	± 25 mV	J1-J65

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	CB WT EN	42	2.4	-0.05	0.05	mA	1-3
K2	CB AD/WT0,14	52					
K3	CB AD/WT1,15	53					
K4	CB AD/WT2,16	54					
K5	CB AD/WT3,17	55					
K6	CB AD/WT4,18	56					
K7	CB AD/WT5,19	57					
K8	CB AD/WT6,20	58					
K9	CB AD/WT7,21	59					
K10	CB AD/WT13	61					
K11	CB AD/WT12	62					
K12	CB AD/WT11	63					
K13	CB AD/WT10	64					
K14	CB AD/WT9	65					
K15	CB AD/WT8	66					
K16	CB FUNCT 0	85					
K17	CB FUNCT 1	86					
K18	CB FUNCT 2	87					
K19	CB FUNCT 3	88					
K20	CB OR	89	2.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
Low level							
K21	CB WT EN	42	0.4	-0.05	0.05	mA	1-3
K22	CB AD/WT0,14	52					
K23	CB AD/WT1,15	53					
K24	CB AD/WT2,16	54					
K25	CB AD/WT3,17	55					
K26	CB AD/WT4,18	56					
K27	CB AD/WT5,19	57					
K28	CB AD/WT6,20	58					
K29	CB AD/WT7,21	59					
K30	CB AD/WT13	61					
K31	CB AD/WT12	62					
K32	CB AD/WT11	63					
K33	CB AD/WT10	64					
K34	CB AD/WT9	65					
K35	CB AD/WT8	66					
K36	CB FUNCT 0	85					
K37	CB FUNCT 1	86					
K38	CB FUNCT 2	87					
K39	CB FUNCT 3	88					
K40	CB OR	89	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.5 V -16.5 V	± 55 mV ± 165 mV	K1-K40 K1-K40
Power supply current limit, (Vcc) (V2)	1.44 A -0.10 A	NA NA	K1-K40 K1-K40
Forcing voltage, (Voh) (Vol)	2.4 V 0.4 V	± 50 mV ± 50 mV	K1-K20 K21-K40
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K40 K1-K40
Output current measurement	NA	± 5 uA	K1-K40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	EN AD/FUNC2D	45	-18.0	-1.2	0.0	V	1-3
~~L2~~	~~EN AD OR WT~~	~~67~~					
L3	EN AD/FUNC	68					
L4	CB DTR	76					
L5	DMA DISABLE	77					
L6	CB DATTMOUT	79					
L7	MACLR	80					
L8	CB ADTMOUT	82					
L9	DTR PHS10/IN	91	-18.0	-1.2	0.0	V	1-3
L10	25.6 uSECCLK	92	-18.0	-1.5	0.0	V	1-3
L11	DMA LPBK	95	-18.0	-1.2	0.0	V	1-3
L12	CB BUSY	96					
L13	CB ATR	97					
L14	CB GRANT	98					
L15	CB CLK	100	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply; (Vcc) (V2)	Off Off	NA NA	L1-L15 L1-L15
Power supply current limit, (Vcc) (V2)	NA NA	NA NA	L1-L15 L1-L15
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L15
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	L1-L15 L1-L15
Input voltage measurement	NA NA NA	± 60 mV ± 75 mV ± 60 mV	L1-L9 L10 L11-L15

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC, Vcc	1	7.4	13.8	uF	1-4
M2	PSFC, Vcc	51	10.4	19.4	uF	1-4
M3	PSFC, V2	50	2.3	4.3	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1-M3
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1-M3
Capacitance measurement	NA NA NA	± 320 nF ± 450 nF ± 100 nF	M1 M2 M3

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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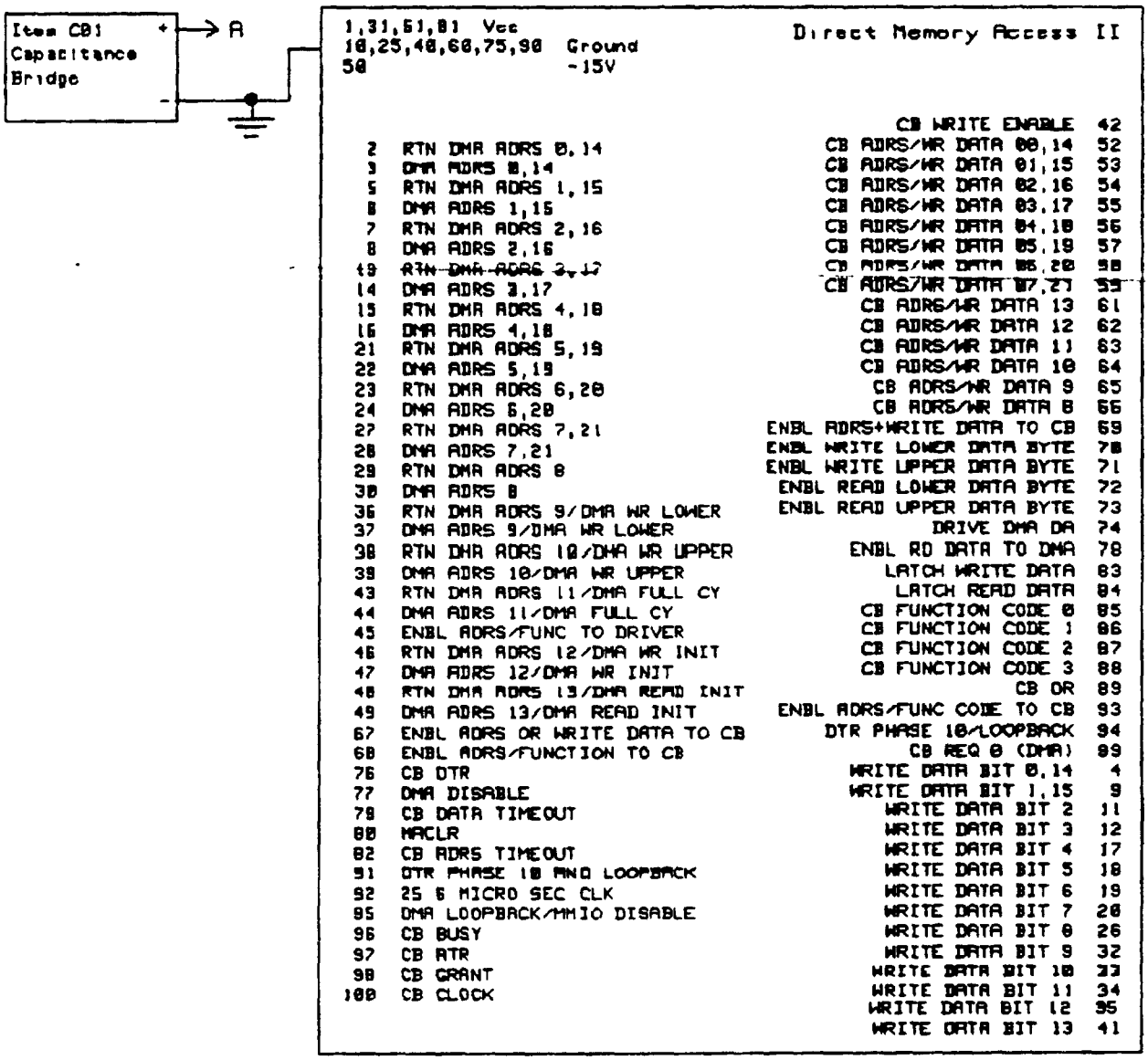


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Icc, Vcc	1,31, 51,81	400	960	mA	1-3
N2	Icc, V2	50	-65	-27	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
(V2)	-16.5 V	± 165 mV	N1,N2
Power supply current limit, (Vcc)	1.44 A	NA	N1,N2
(V2)	-0.10 A	NA	N1,N2
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1,N2
low, (VL)	0.0 V	± 100 mV	N1,N2
Power supply current measurement	NA	± 28.0 mA	N1
	NA	± 1.9 mA	N2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (1500 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (1500 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	P1
(V2)	-15.0 V	± 150 mV	P1,P2
(Vcc)	5.5 V	± 55 mV	P2
Power supply current limit, (Vcc)	1.44 A	NA	P1,P2
(V2)	-0.10 A	NA	P1,P2
Pattern voltage high, (VH)	2.4 V	± 100 mV	P1,P2
low, (VL)	0.4 V	± 100 mV	P1,P2
Output voltage verification high, (Voh)	NA	± 105 mV	P1
high, (Voh)	NA	± 155 mV	P2
low, (Vol)	NA	± 25 mV	P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (1500 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (1500 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (1500 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (1500 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2) (Vcc)	4.5 V -15.0 V 5.5 V	± 45 mV ± 150 mV ± 55 mV	Q1,Q3 Q1-Q4 Q2,Q4
Power supply current limit, (Vcc) (V2)	1.44 A -0.10 A	NA NA	Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 100 mV	Q1-Q4
low, (VL)	0.4 V	± 100 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

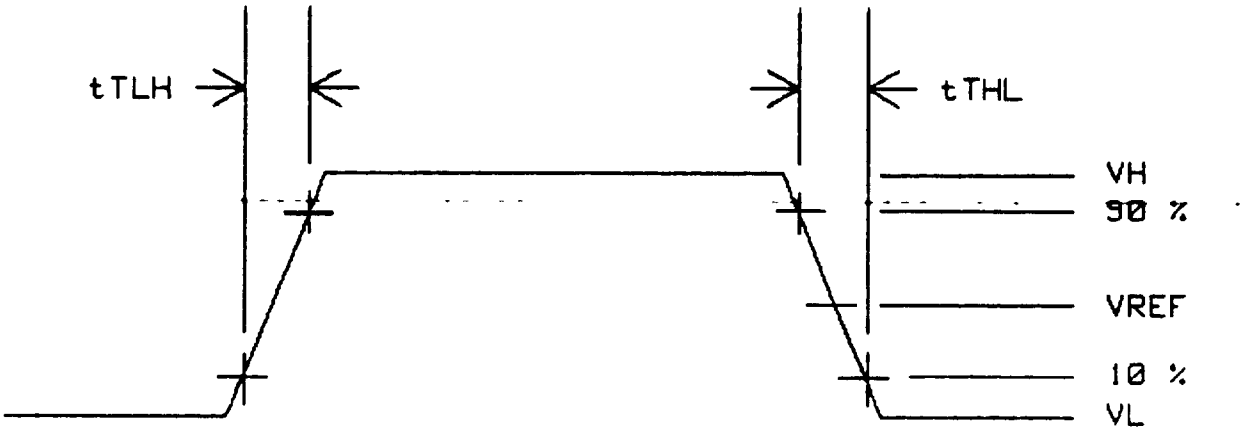
(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:

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NOTES: - Continued.



Where:

(a) VH and VL are as specified in note 2 (b).

(b) tTLH = tTHL ≤ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to CB OR	-	-	-	36	36	36	nS	11
2	tpd	CB CLK to EN AD+WT DAT	-	-	-	38	38	38	nS	11
3	tpd	CB CLK to DTR PHS10	-	-	-	37	37	37	nS	11
4	tpd	CB CLK to EN AD/FUNC2C	-	-	-	38	38	38	nS	11
5	tpd	CB CLK to LTCH RD DAT	-	-	-	41	41	41	nS	11
6	tpd	CB CLK to DRIVE DMA	-	-	-	47	47	47	nS	11
7	tpd	CB CLK to EN WT DATLWR	-	-	-	48	48	48	nS	11
7	tpd	CB CLK to EN WT DATUPR	-	-	-	48	48	48	nS	11
8	tpd	CB CLK to EN RD DATLWR	-	-	-	48	48	48	nS	11
8	tpd	CB CLK to EN RD DATUPR	-	-	-	48	48	48	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	CB CLK to LTCH WT DAT	-	-	-	55	55	55	nS	11
10	tpd	CB CLK to CB WT EN	-	-	-	62	62	62	nS	11
11	tpd	DMA AD 0,14 to CB AD/WT0,14	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 1,15 to CB AD/WT1,15	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 2,16 to CB AD/WT2,16	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 3,17 to CB AD/WT3,17	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 4,18 to CB AD/WT4,18	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 5,19 to CB AD/WT5,19	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 6,20 to CB AD/WT6,20	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 7,21 to CB AD/WT7,21	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 8 to CB AD/WT8	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 9 to CB AD/WT9	-	-	-	91	91	91	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	DMA AD 10 to CB AD/WT10	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 11 to CB AD/WT11	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 12 to CB AD/WT12	-	-	-	91	91	91	nS	11
11	tpd	DMA AD 13 to CB AD/WT13	-	-	-	91	91	91	nS	11

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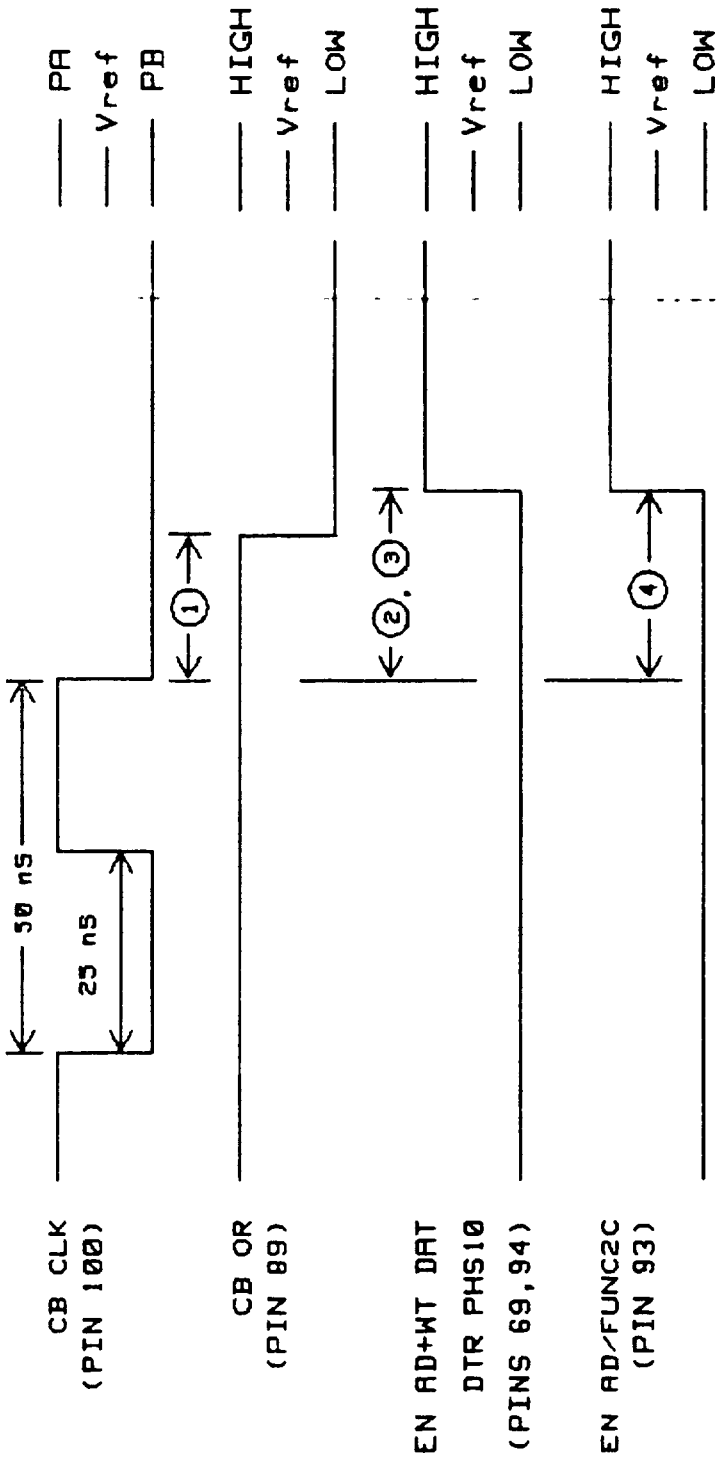


FIGURE 11a. Dynamic waveforms for table XVII.

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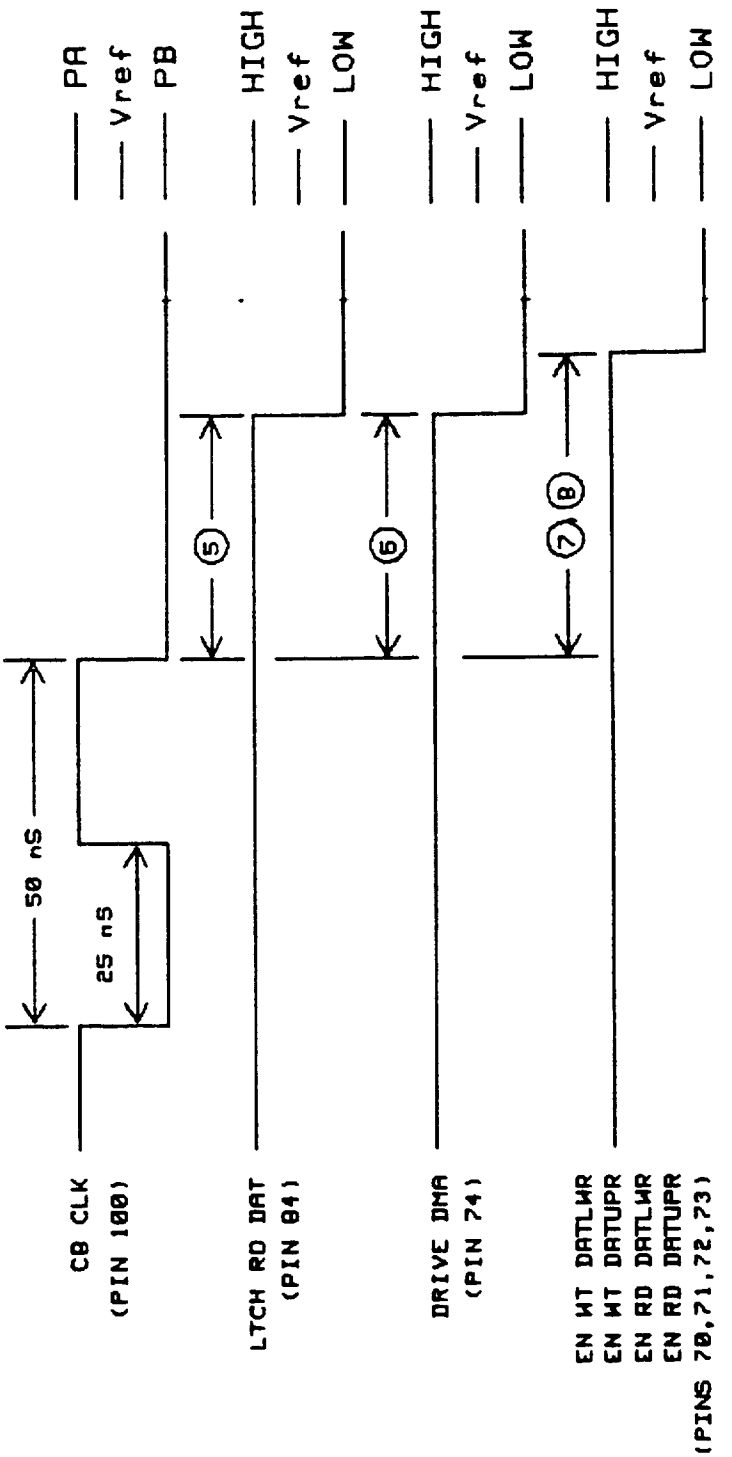


FIGURE 11b. Dynamic waveforms for table XVII.

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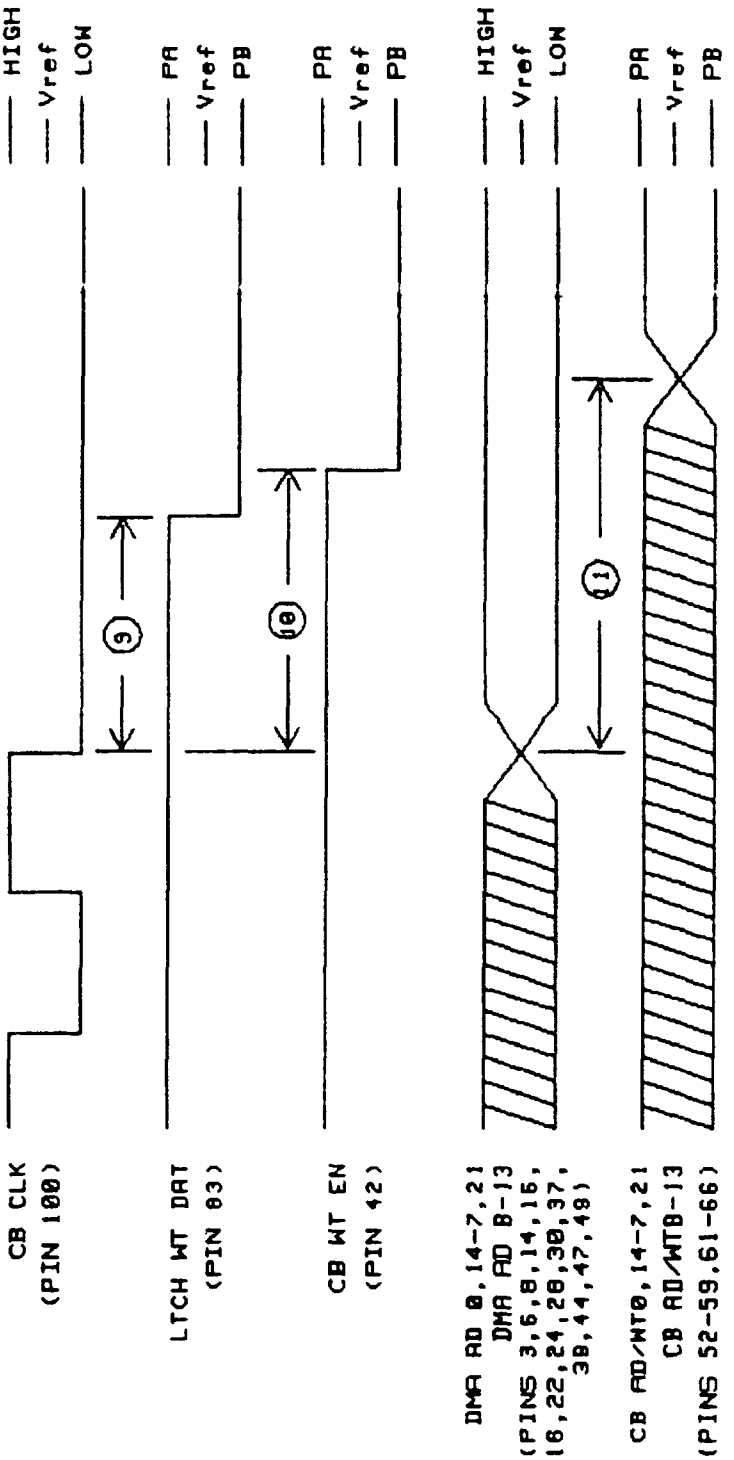


FIGURE 11c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	RTN DMA 0,14	2	4	50	pF	1-4
R2	DMA AD 0,14	3	4	75	pF	1-4
R3	WT DAT 0,14	4	8	100	pF	1-4
R4	RTN DMA 1,15	5	4	75	pF	1-4
R5	DMA AD 1,15	6	4	75	pF	1-4
R6	RTN DMA 2,16	7	4	50	pF	1-4
R7	DMA AD 2,16	8	4	75	pF	1-4
R8	WT DAT 1,15	9	8	125	pF	1-4
R9	WT DAT 2	11	10	125	pF	1-4
R10	WT DAT 3	12	10	125	pF	1-4
R11	RTN DMA 3,17	13	4	75	pF	1-4
R12	DMA AD 3,17	14				
R13	RTN DMA 4,18	15				
R14	DMA AD 4,18	16	4	75	pF	1-4
R15	WT DAT 4	17	10	125	pF	1-4
R16	WT DAT 5	18	10	125	pF	1-4
R17	WT DAT 6	19	10	150	pF	1-4
R18	WT DAT 7	20	10	125	pF	1-4
R19	RTN DMA 5,19	21	4	75	pF	1-4
R20	DMA AD 5,19	22	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	RTN DMA 6,20	23	4	50	pF	1-4
R22	DMA AD 6,20	24	4	75	pF	1-4
R23	WT DAT 8	26	8	125	pF	1-4
R24	RTN DMA 7,21	27	4	75	pF	1-4
R25	DMA AD 7,21	28	4	75	pF	1-4
R26	RTN DMA 8	29	4	50	pF	1-4
R27	DMA AD 8	30	4	75	pF	1-4
R28	WT DAT 9	32	18	200	pF	1-4
R29	WT DAT 10	33	16	200	pF	1-4
R30	WT DAT 11	34	14	175	pF	1-4
R31	WT DAT 12	35	10	150	pF	1-4
R32	RTN DMA 9	36	4	75	pF	1-4
R33	DMA AD 9	37	4	75	pF	1-4
R34	RTN DMA 10	38	4	50	pF	1-4
R35	DMA AD 10	39	4	75	pF	1-4
R36	WT DAT 13	41	10	125	pF	1-4
R37	RTN DMA 11	43	4	75	pF	1-4
R38	DMA AD 11	44	4	50	pF	1-4
R39	EN AD/FUNC2D	45	4	100	pF	1-4
R40	RTN DMA 12	46	4	50	pF	1-4

TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	DMA AD 12	47	4	75	pF	1-4
R42	RTN DMA 13	48	4	75	pF	1-4
R43	DMA AD 13	49				
R44	EN AD OR WT	67	10	125	pF	1-4
R45	EN AD/FUNC	68	10	125	pF	1-4
R46	DMA DISABLE	77	2	125	pF	1-4
R47	CB DATTMOUT	79	2	75	pF	1-4
R48	MACLR	80	6	75	pF	1-4
R49	CB ADTMOUT	82	2	50	pF	1-4
R50	DTR PHS10/IN	91	2	75	pF	1-4
R51	25.6 uSECCLK	92	2	50	pF	1-4
R52	DMA LPBK	95	2	75	pF	1-4
R53	CB WT EN	42				
R54	CB AD/WT0,14	52				
R55	CB AD/WT1,15	53				
R56	CB AD/WT2,16	54				
R57	CB AD/WT3,17	55				
R58	CB AD/WT4,18	56				
R59	CB AD/WT5,19	57				
R60	CB AD/WT6,20	58	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	CB AD/WT7,21	59	2	75	pF	1-4
R62	CB AD/WT13	61				
R63	CB AD/WT12	62				
R64	CB AD/WT11	63				
R65	CB AD/WT10	64				
R66	CB AD/WT9	65				
R67	CB AD/WT8	66				
R68	CB FUNCT 0	85				
R69	CB FUNCT 1	86				
R70	CB FUNCT 2	87				
R71	CB FUNCT 3	88				
R72	CB OR	89	2	75	pF	1-4
R73	CB DTR	76	2	50	pF	1-4
R74	CB BUSY	96	4	75	pF	1-4
R75	CB ATR	97	2	50	pF	1-4
R76	CB GRANT	98	6	100	pF	1-4
R77	CB CLK	100	2	50	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R77
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R77
Capacitance measurement	NA	± 2.3 pF	R1
	NA	± 3.55 pF	R2
	NA	± 4.6 pF	R3
	NA	± 3.55 pF	R4,R5
	NA	± 2.6 pF	R6
	NA	± 3.55 pF	R7
	NA	± 5.85 pF	R8
	NA	± 5.75 pF	R9,R10
	NA	± 3.55 pF	R11-R14
	NA	± 5.75 pF	R15,R16
	NA	± 7.0 pF	R17
	NA	± 5.75 pF	R18
	NA	± 3.55 pF	R19,R20
	NA	± 2.3 pF	R21
	NA	± 3.55 pF	R22
	NA	± 5.85 pF	R23
	NA	± 3.55 pF	R24,R25
	NA	± 2.3 pF	R26
	NA	± 3.55 pF	R27
	NA	± 9.1 pF	R28
	NA	± 9.2 pF	R29
	NA	± 8.05 pF	R30
	NA	± 7.0 pF	R31
	NA	± 3.55 pF	R32,R33
	NA	± 2.3 pF	R34
	NA	± 3.55 pF	R35
	NA	± 5.75 pF	R36
	NA	± 3.55 pF	R37
	NA	± 2.3 pF	R38
	NA	± 4.8 pF	R39

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 2.3 pF	R40
	NA	± 3.55 pF	R41-R43
	NA	± 5.75 pF	R44,R45
	NA	± 6.15 pF	R46
	NA	± 3.65 pF	R47
	NA	± 3.45 pF	R48
	NA	± 2.4 pF	R49
	NA	± 3.65 pF	R50
	NA	± 2.4 pF	R51
	NA	± 3.65 pF	R52-R72
	NA	± 2.4 pF	R73
	NA	± 3.55 pF	R74
	NA	± 2.4 pF	R75
	NA	± 4.7 pF	R76
	NA	± 2.4 pF	R77

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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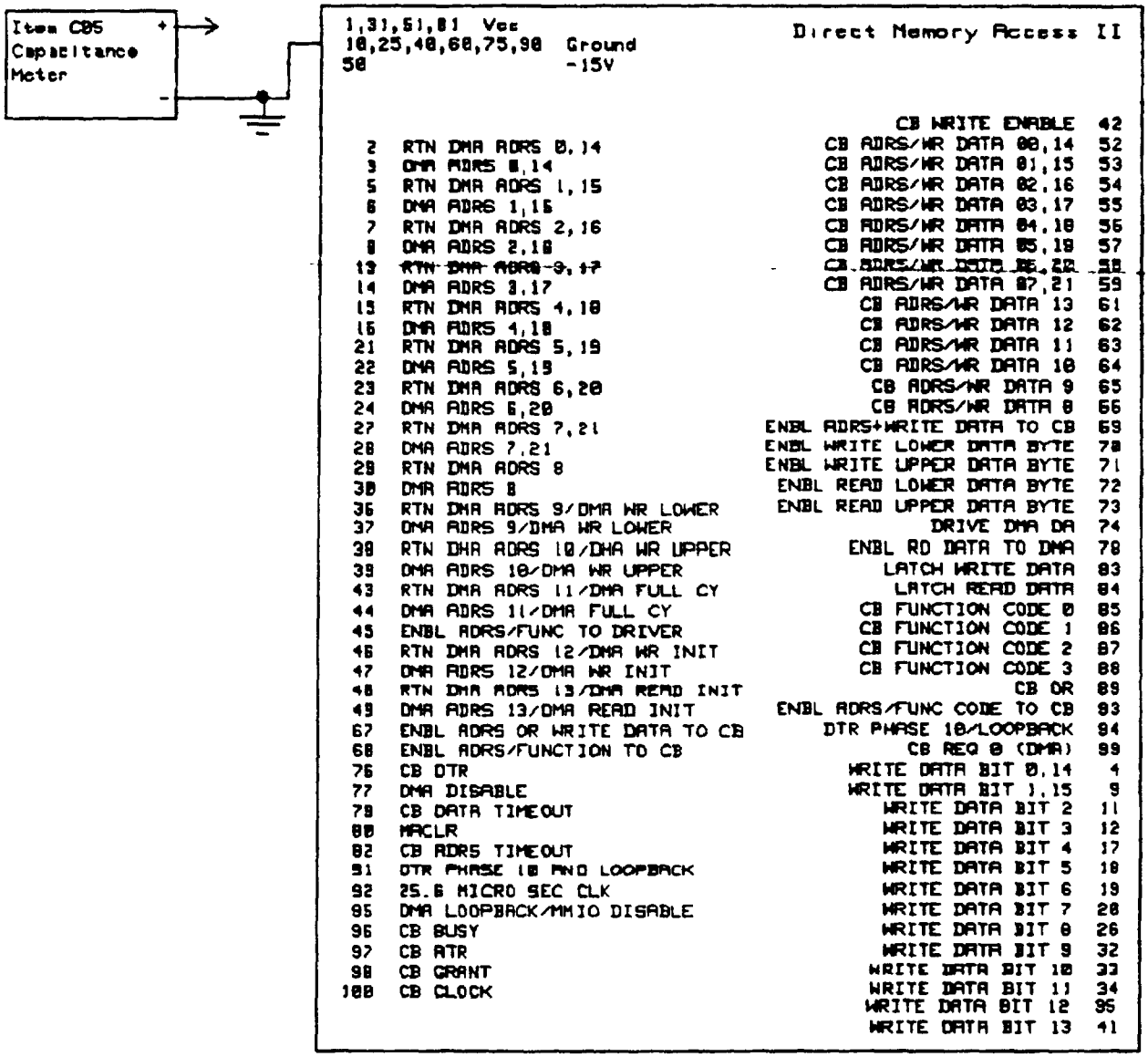


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latchup test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S	MACLR	80	-18.0	-1.2	2.4	V	1-4

NOTES:

1. Input clamp diode latchup test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	4.5 V -15.0 V	± 45 mV ± 150 mV	S S
Power supply current limit, (Vcc) (V2)	1.44 A -0.10 A	NA NA	S S
Forcing current, (Iin)	-18 mA	± 1.8 mA	S
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	S S
Input voltage measurement	NA	± 180 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

3. Apply input pulse for a duration of 500 ± 20 nS.

4. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA, Vcc	1	0	400	mV	1-4
T2	PSTA, Vcc	51	0	400	mV	1-4
T3	PSTA, V2	50	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.0 V -15.0 V	± 250 mV ± 750 mV	T1-T3 T1-T3
Power supply current limit, (Vcc) (V2)	1.44 A -0.10 A	NA NA	T1-T3 T1-T3
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	T1-T3 T1-T3
Power supply transient amplitude measurement	NA	± 20 mV	T1-T3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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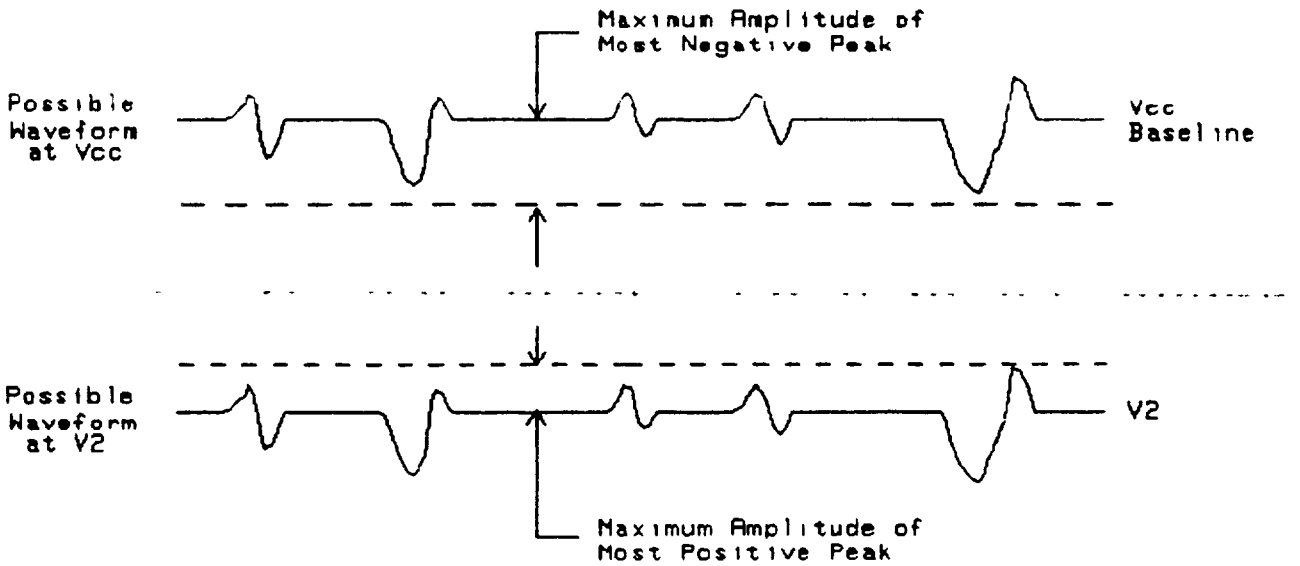
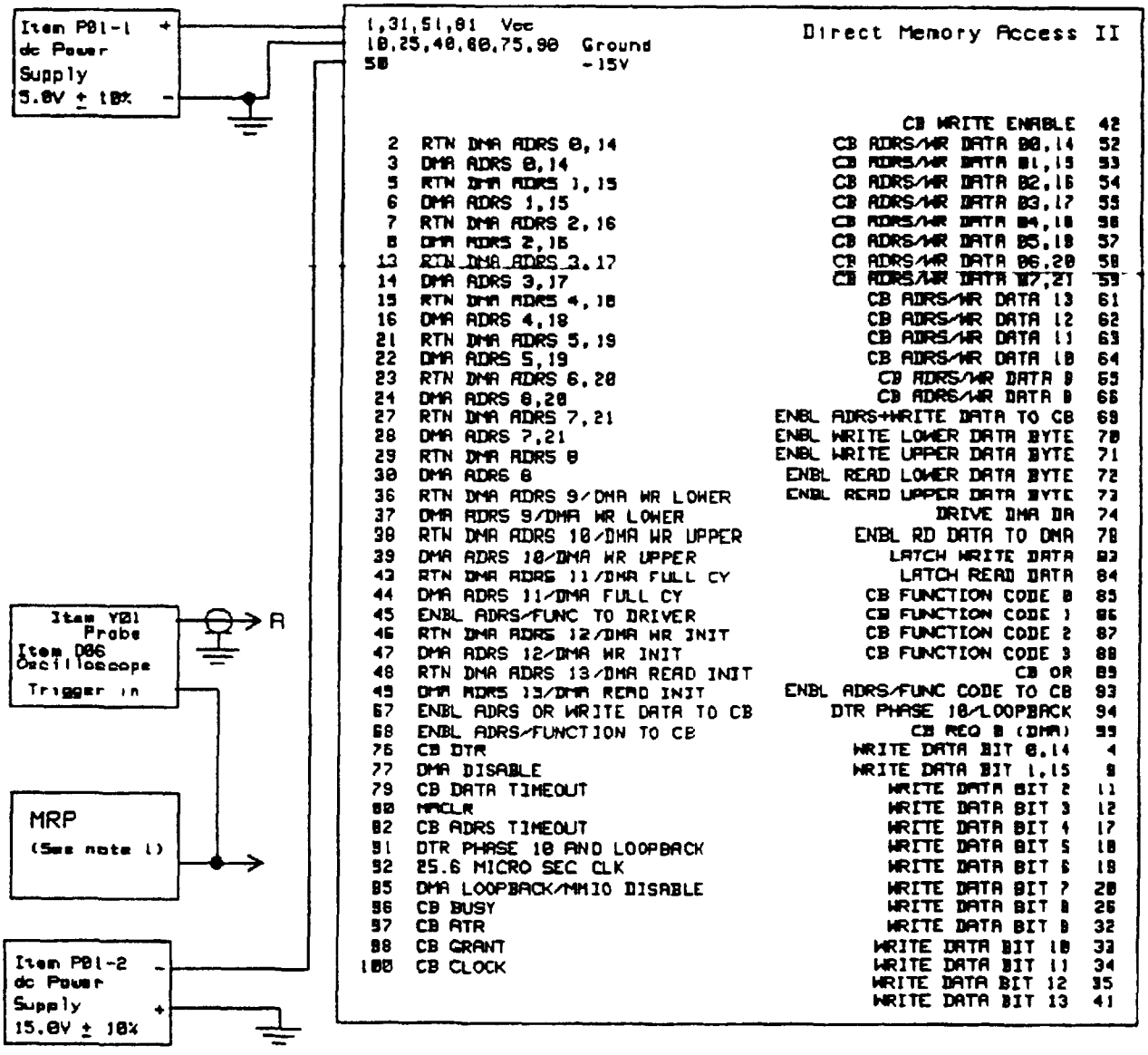


FIGURE 13. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-28787/527	Modules, Standard Electronic, Direct Memory Access II Module, Key Code DWH

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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